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Federated learning at the edge in Industrial Internet of Things: A review

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ABSTRACT

The convergence of Federated learning (FL) and Edge computing (EC) has emerged as an essential paradigm, particularly within the Industrial Internet of Things (IIoT) to enable the intelligent decision making. This work diligently examines the current state-of-the-art research at the intersection of FL, EC, and IIoT. An extensive review of the literature explores the diverse applications and challenges associated with this integration. The challenges range from privacy preservation and communication overhead to resource allocation. The incorporation of edge devices at which ensuring the federated learning in distributed manner helps to minimize energy consumption in IIoT, ultimately leads to a sustainable computing environment. By exploring the existing literature and research advancements, our goal is to highlight existing Edge-IoT software and hardware platforms and assess their usability in addressing challenges. In addition, we review existing recent frameworks, methodologies, and models employed to address these challenges, focusing on key performance matrices and its domain such as application, networking, and learning. We highlight the achievements and potential of FL and EC and underscore the need for tailored solutions to suit the unique demands of IIoT. Furthermore, we identify some of the major challenges as opportunities for future research, requires interdisciplinary collaboration and innovative algorithmic solutions. This work can help navigate through the challenges and unlock the full potential, contributing to the advancement of future IIoT applications.

1. Introduction

The shift from traditional manufacturing to a digital paradigm is driven by significant advancement in computing and communication technologies [1,2]. Cutting-edge technologies are leading this significant shift, with key entities such as Micro-electro-mechanical systems (MEMS), advanced sensor technologies, the latest communication systems (4G, 5G, and 6G), artificial intelligence (AI), collaborative robotics, cloud computing, network virtualization (NV), and software-defined networks (SDNs). Adapting from Industry 4.0 to 5.0 has been beneficial in boosting the working environment, productivity, and profitability. In the domain of IIoT, the integration of machine learning (ML) algorithms has emerged as a critical time-sensitive adoption because it enables intelligent decision-making and predictive capabilities. However, the conventional cloud-based approaches for handling ML in IIoT environments have inherent challenges. Current practices often involve transmitting vast amount of raw sensor data from devices to remote cloud servers for processing, where ML models are trained and predictions are made. While this centralized model has yielded significant insights, it introduces high latency due to increased amount of data and its transmission (bandwidth) cost [3]. Additionally, the concerns about data sharing over the cloud due to the security and privacy of the centralized learning make it less suitable for sensitive applications requiring real-time decision-making [4,5]. The recent approaches to handle these challenges involved decentralized systems (data, computation, and learning) such as Edge computing (EC) [6] and Federated learning (FL) [7–9] to substitute the Cloud computing and ML.

The adoption of FL [7–9] and EC [10,11] in the IIoT is driven by several compelling technical aspects. It enables collaborative model training across decentralized ED while preserving data privacy by keeping sensitive information localized [12]. This distributed learning paradigm is particularly beneficial in IIoT because data generation and storage often happen on the Edge. Additionally, preventing massive data transfers to a centralized server can help mitigate bandwidth constraints in IIoT networks; EC brings computational capabilities closer to the data source. This can significantly benefit time-sensitive, real-time processing and low latency applications. In FL, computing power can be distributed over multiple EDs, following the collaborative model training. Running FL on EDs can enable the training as a shared model without central aggregation of data which ca reduced latency, enhanced privacy protection along with scalability of the overall system.

The proliferation of ED has driven the shift towards industry automation and created a demand for intelligent decision making at

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the edge of the network [13], and FL emerges as a promising approach. This work explores the existing Edge-based paradigms for handling FL in IIoT, highlighting their problem statements, methodologies, strengths, results benchmarks, and limitations. Additionally, emerging strategies that leverage FL and EC to address the evolving needs of dynamic ecosystems are also discussed. This work aims to provide a comprehensive overview of the FL, ED, methodologies, and their architecture within the context of IIoT and the Table 1, describe the abbreviation used in the work is provided.

FL addresses key challenges that traditional centralized Machine Learning (ML) faces in IIoT environments, particularly regarding privacy, bandwidth, and latency. In IIoT applications, data generated at the edge often includes sensitive operational information (e.g., industrial process data or proprietary machine diagnostics), making decentralized learning crucial for privacy preservation. Moreover, transferring large amount of raw data to a centralized server for processing is not feasible in IIoT, given the limited bandwidth and stringent latency requirements. Studies have shown that FL minimizes these challenges by keeping data localized [8,9]. This decentralized model allows edge devices to collaboratively train a global model without extensive data transfer, preserving both data privacy and reducing network strain.

IIoT applications often involve real-time monitoring, predictive maintenance, and safety-critical processes, where even slight delays or data loss can result in significant disruptions or safety hazards. Unlike consumer IoT devices, IIoT systems frequently operate in low-bandwidth environments, making the transmission of large model updates challenging. To address these constraints, FL in IIoT must incorporate techniques such as model compression, gradient sparsification, and efficient aggregation protocols [14,15], balancing real-time responsiveness with communication efficiency.

Our work makes several distinct contributions to advancing FL within the context of IIoT, compared to existing literature. Specifically, we provide a comprehensive review that not only addresses the architectural and technical integration of FL and edge computing but also emphasizes the practical constraints unique to IIoT environments, such as real-time latency requirements, limited computational resources, and enhanced security needs. By examining the intersection of these domains, our survey highlights previously unaddressed challenges, proposes future research directions, and outlines potential applications that leverage FL in IIoT for improved privacy, scalability, and system robustness.

1.1. Research questions and objectives

To structure our review and guide readers through the main issues in FL for IIoT, we outline the following research questions:

- What unique challenges does FL face in the IIoT context, especially regarding latency, bandwidth, and security concerns?
- How can FL be adapted to meet the scalability and communication efficiency requirements specific to IIoT?
- What key technical frameworks and methodologies have been proposed to support privacy-preserving, efficient, and robust FL in IIoT?
- What open issues and future research directions should be prioritized to realize the full potential of FL in industrial applications?

This review aims to answer these questions, providing insights into the advancements, limitations, and future directions for FL in IIoT environments.

The organization of the paper is as follows: Section 2 provides a review of related work, focusing on prior surveys of FL and EC in IIoT contexts. Section 3 introduces the fundamentals of FL, followed by a detailed overview of important edge devices and platforms in Section 4. Section 5 explores technical challenges and solution techniques across various IIoT use cases, while Section 6 identifies open issues and future research directions. Finally, we conclude in Section 7 with a summary and potential applications.

Table 1
List of abbreviations.

Abbreviation	Definition
IIoT	Industrial Internet of Things
FL	Federated Learning
FCN	Fully connected layers
EC	Edge Computing
ED	Edge Device
PP	Privacy Preservation
QoS	Quality of Service
HFL	Hierarchical Federated Learning
AFL	Asynchronous Federated Learning
PFL	Personalized Federated Learning
DP	Differential Privacy
SA	Secure Aggregation
MQTT	Message Queuing Telemetry Transport
CoAP	Constrained Application Protocol
TSN	Time-Sensitive Networking
FedProx	Federated Proximal (an FL algorithm for handling heterogeneity)
DT	Digital Twin
AD	Anomaly Detection
ID	Intrusion Detection
FedAvg	Federated Averaging
MDP	Markov Decision Process
DRL	Deep Reinforcement Learning
CNN	Convolutional Neural Network
TPU	Tensor Processing Unit
VPU	Vision Processing Unit
FPGA	Field-Programmable Gate Array
UAV	Unmanned Aerial Vehicle
FedDist	Federated Distillation

2. Related work

The landscape of FL and EC has gained significant attention in recent surveys, in the context of IoT or IIoT. However, most of these studies have traditionally treated FL and EC as separate domains, often neglecting the critical considerations of IIoT inclusion diligently. In the domain of Edge intelligence for IIoT networks, the benefit of data fusion is introduced in [11], while [10] has explored offloading computational capacity on mobile EC for IoT networks. The network automation for IIoT and applications has been explored in [1,2] which has examined advanced manufacturing computational offloading modeling for EC encompassing communication, computational, and energy models for IIoT systems. Some related works have utilized FL algorithms within communication protocols, employing optimization algorithms. For instance, [13] provided an overview of EC, FL and digital twins for smart cities, while [6] gave insight into the future of EC provisioning. Considering user privacy, [7] assessed differential privacy mechanisms in deep federated learning. Additionally, [9] has comprehensively explored economic and game theoretic motivations for data owners to engage in FL training for industrial applications. In [7–9], it considered the use of FL algorithms in different applications, system platforms, real-world use cases, and contextual processes. Blockchain's integration with EC applications [4,5] has also surfaced; however, blockchain protocol is very slow due to the maintenance of its ledger in distributed environment; and not very often used in IIoT. The deep learning algorithm for the IoT network is also explored for task offloading [16,17] and task scheduling [18] though its has used architecture such as cloud and fog computing.

To conduct this review, we have performed the analytical search for paper extraction using the Scopus database[1]. As shown in Fig. 1, very limited surveys have been published in peer-reviewed journals and conferences. The title search is performed on Scopus using different search queries and the number of articles (#) available for that given search query is also shown. The literature search is performed with

[1] https://www.scopus.com/

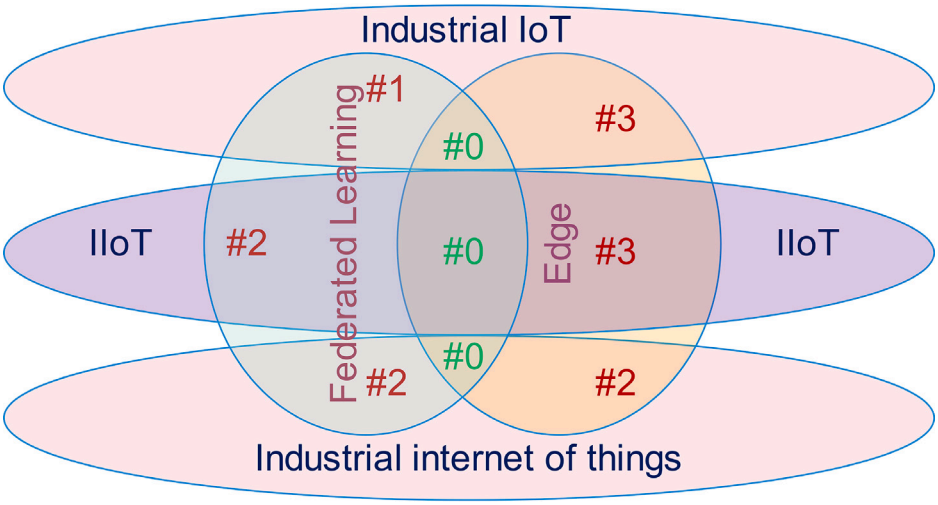
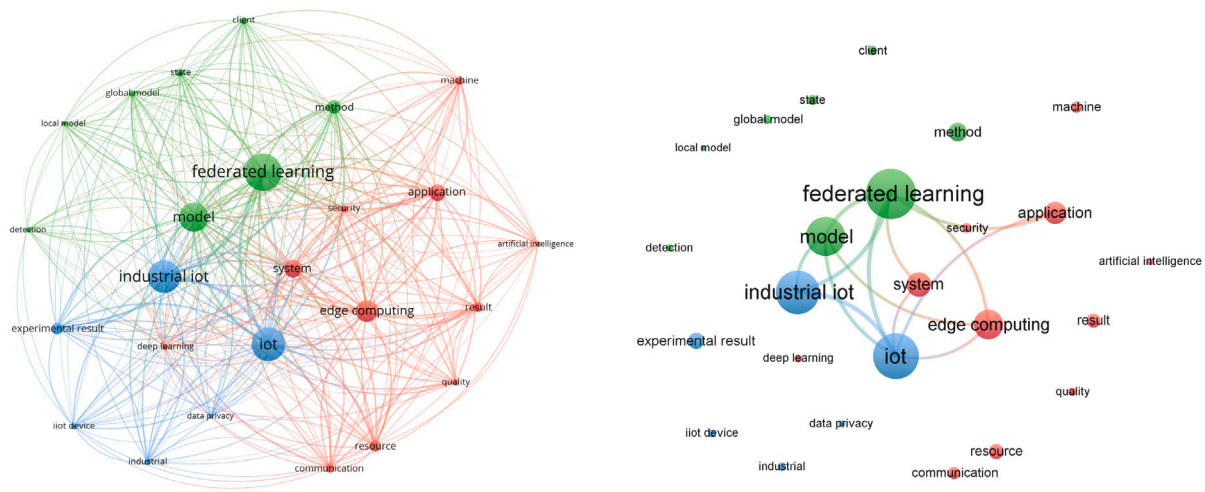


Fig. 1. Number of existing survey/review papers on Edge computing or federated learning in industrial IoT domain.



(a) Comprehensive network visualization of key terms and their relationships in Industrial IoT research

(b) Simplified abstraction of dominant connections in the network of IIoT-related concepts

Fig. 2. Relationships in federated learning, edge computing and Industrial IoT research.

few refinements as in the first phase, we quickly reviewed the search string without any filter where "edge OR federated learning" has been common along with IoT and IIoT. Interestingly, no comprehensive review exists for all the strings which is reported in Fig. 1. Later, we employed the review article filter and used different search strings to cover the literature deals with road-map, future prospective, challenge, and vision. As shown in Fig. 1, there are no survey or review papers addressing federated learning on EDs in industrial IoT, which is the main focus of our work.

The provided Fig. 2 highlight the interconnectedness of key research areas in the field of Industrial IoT (IIoT). In our .bib file, there are mainly three cluster is being formed as you as observe from shown diagram. In Fig. 2(a) illustrates a comprehensive network visualization showcasing the relationship between federated learning, edge computing, and IoT in IIoT applications. Fig. 2(b) presents a simplified abstraction of this network, focusing on the stronger, more direct connections between these concepts. Both figures emphasize the survey's focus on the synergy between federated learning and edge computing in improving security, communication efficiency, and data privacy within IIoT frameworks.

3. Fundamentals of federated learning

FL is an emerging paradigm that enables collaborative model training across distributed devices while preserving data privacy. Unlike traditional centralized learning, FL ensures that raw data remains on edge devices, and only model updates are shared with a central server. This decentralized approach addresses critical challenges in Industrial Internet of Things (IIoT) systems, such as data privacy, communication efficiency, and computational scalability. This section delves into the core principles of FL, emphasizing its architecture, training mechanisms, and the unique requirements posed by IIoT environments. The discussion includes key concepts such as model aggregation techniques, hierarchical FL, and the implications of heterogeneous data and resources across edge devices.

In scenarios, where data is sensitive, "in-large" volume, or subject to privacy regulations, centralizing the data may be infeasible or undesirable. The local data is being used for training (storage) in the ED while the central server collaborates in the training by sharing the initial model along with its parameters with the ED as shown in Fig. 3 [19]. Afterward, the server collects the updated model parameters that have been trained locally. Combining these updated

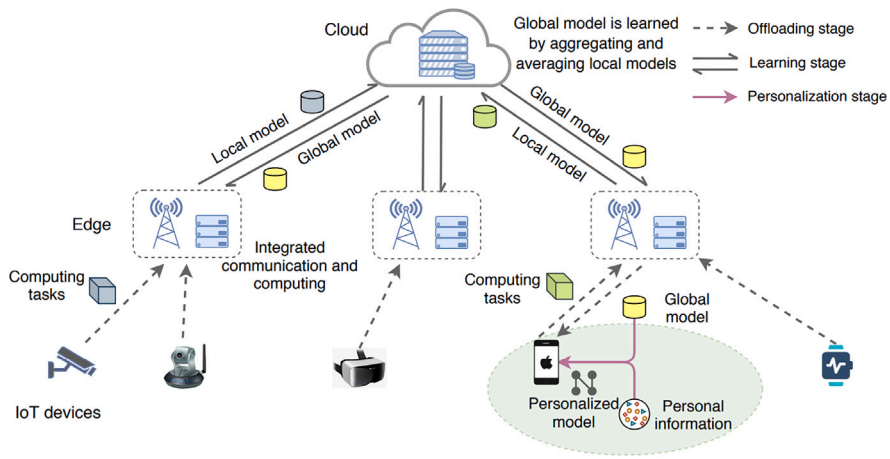


Fig. 3. Architecture of federated learning over cloud–Edge platform.

models can help generate a newer global model, achieving convergence through iterations. In exploring FL on EDs, various approaches contribute to its efficient implementation [9]. One such strategy is Hierarchical FL, where EDs are organized into hierarchical groups, allowing local model training within each group and subsequent aggregation to form a global model [15,20,21]. This hierarchical structure significantly reduces communication overhead and computational burden, making it suitable for large-scale deployments. Vertical FL focuses on scenarios where data is vertically partitioned across devices, with each device holding a distinct set of features. This approach ensures collaborative model training while preserving data privacy, making it ideal for situations where confidentiality is crucial. In contrast, horizontal FL addresses horizontally partitioned data, allowing EDs with independent data samples to collaboratively train a global model [22]. Optimization algorithms like "Federated Averaging" and communication protocols play vital roles in effective FL. These protocols facilitate secure and efficient parameter exchange, ensuring the transmission of model updates while maintaining data privacy and minimizing communication overhead.

Another important aspect is privacy preservation. Since the local data is used for training in EDs, they must ensure that sensitive information is not exposed. Techniques such as secure aggregation, differential privacy, and encryption methods are often used to protect the privacy of individual data samples and model updates [12]. Distributing data across EDs has been a critical challenge in FL because data is often non-independent and identically distributed, meaning that each ED may have a different data distribution. This can pose challenges in achieving model convergence due to the heterogeneity of local datasets. To address this, federated learning employs various techniques such as federated averaging, model personalization, and adaptive learning algorithms to account for the differences in data distribution across EDs.

These frameworks highlight the need for IIoT-specific adaptations of FL to address limitations based on application context. In the next section, we will explore specific EDs that can be useful in implementing FL in industrial automation.

4. Edge devices and platforms in industrial automation

This section provides examples of some major Edge devices and platforms widely used in the industrial automation.

4.1. Edge devices (EDs)

The EDs are specialized hardware accelerators developed by multiple vendors; specifically designed to perform inference for machine learning models locally (geographically close to IoT devices), without relying on cloud-based solutions [23]. Edge Tensor Processing Units (TPUs), like other similar devices, are part of the broader category of hardware accelerators tailored for EC and AI inference. These EDs are specifically optimized for executing lighter version of models such as TensorFlow Lite models, which are lightweight versions of TensorFlow models suitable for mobile and embedded devices. These devices are engineered to enhance the efficiency and performance of AI tasks at the Edge of the network, close to where data is generated. They are designed to process data locally, reducing latency, improving privacy, and minimizing the need for constant Internet connectivity. Here, we briefly describe some of the major EDs and their main features, listed in Table 2.

1. *Edge TPU*: It is an application-specific integrated circuit (ASIC) that specializes in accelerating ML inference tasks. It is designed to process and execute TensorFlow Lite models, which are optimized versions of ML models suitable for EDs. It offers both high performance and energy efficiency, suitable for various Edge AI applications. It is tightly integrated with TensorFlow Lite and supports various neural network architectures. It comes in different form factors, including USB accelerators and modules that can be integrated into devices.
2. *Coral TPU*: It has been developed by Google as part of the Coral platform, and designed to accelerate ML inference on ED. It offers both USB accelerators and modules that integrate with devices. It is optimized for TensorFlow Lite models and provides hardware acceleration for AI tasks.
3. *NVIDIA Jetson Series*: It consists of embedded computing platforms equipped with powerful Graphics Processing Units (GPUs) and other hardware components optimized for AI workloads. These platforms run on NVIDIA's CUDA programming framework, enabling developers to deploy complex deep-learning models. They are designed for tasks like computer vision, robotics, and autonomous systems. The Jetson devices include the Jetson Nano, Jetson TX2, and Jetson Xavier, each with varying levels of performance and capabilities.
4. *Intel Movidius Vision Processing Unit (VPU)*: It is designed to accelerate vision-related tasks in an ED. It is a low-power hardware accelerator capable of running neural networks with optimized performance and energy efficiency. It is often used in applications, where real-time processing of visual data is crucial, such as surveillance cameras and drones.
5. *Field-Programmable Gate Arrays (FPGAs)*: These are re-configurable hardware devices that can be programmed to perform specific tasks, including AI inference. It offer flexibility to implement custom neural network architectures and optimizations. They can be integrated into EDs to accelerate AI workloads while allowing customization based on specific application needs.

Table 2
Listing some of the existing Edge devised, their application usability or performance scenario, Energy consumption (high, moderate and variable), and Cost per unit.

Device	Performance	Energy consumption	Cost
Edge TPU	Image recognition, NLP	High	High
NVIDIA Jetson	Robotics, Computer Vision	High	Moderate
Intel Movidius	Video Analysis, IoT	Moderate	Moderate
Coral TPU	Image processing, Edge AI	Moderate	High
FPGAs	Custom AI Workloads	Variable	Variable
Xilinx AI Edge	Custom AI Workloads	Variable	Variable
Dell PowerEdge R740	Data Analytics, Virtualization	High	High
HPE ProLiant DL380	Data Management, Virtualization	High	High
Lenovo SR650	Enterprise Applications	High	Moderate
Supermicro SYS-5019S-L	Small-scale Servers, Cloud Services	Moderate	High

6. *Xilinx AI Edge*: It offers adaptable AI-enabled ED that combines FPGA technology with AI capabilities. These devices can be programmed to accelerate AI workloads and handle a range of tasks, from image and speech recognition to data analytics, and provide a balance between performance, flexibility, and power efficiency.
7. *Dell PowerEdge R740*: It is a versatile ED with a general-purpose focus that supports custom implementations and is known for its high performance in data analytics and virtualization tasks. With its robust capabilities, it caters to applications that require substantial computational resources and efficient data management.
8. *HPE ProLiant DL380 Gen10*: It is another general-purpose ED that accommodates custom implementations. Its high performance is well-suited for various tasks, particularly data management and virtualization. It can serve as a reliable solution for enterprises seeking advanced processing capabilities.
9. *Lenovo ThinkSystem SR650*: It is a general-purpose ED tailored for custom implementations. Its high performance makes it an ideal choice for a range of applications, particularly those associated with enterprise environments. It is equipped to handle diverse enterprise applications with efficiency.
10. *Supermicro SYS-5019S-L*: It is designed as a general-purpose ED with a focus on custom implementations to offer a balanced performance profile, making it suitable for small-scale servers and cloud services. It caters to scenarios where moderate performance coupled with high reliability is essential.

4.2. Edge platforms

The adoption of FL and the IIoT has given rise to a diverse landscape of platforms tailored to address the unique demands of data processing at the Edge. A comprehensive comparison reveals multiple Edge-Cloud platforms, each with distinct ownership models and key features. We have listed some of the main Edge-Cloud platforms with its features and suitability in Table 3. The Google Cloud IoT platform combines cloud services (including ML) like Bigtable with Linux kernel integration, despite limited access during the trial phase. In contrast, Microsoft introduced IoT Edge and IoT Centralized on its Azure platform, offering compatibility with both Linux and Windows environments, while operating on a paying pro-rata basis. Similarly, the IoT platform of AWS seamlessly integrates Greengrass and FreeRTOS, catering to a variety of hardware platforms through the AWS Partner Network, under a flexible pay-per-usage framework. Meanwhile, IBM's Watson IoT and Bosch IoT Suite extend their proprietary cloud offerings with features like Watson API and Bosch IoT Remote Manager, targeting Linux, Windows, and Bosch hardware, respectively. Siemens Mindsphere focuses on open flexibility, providing MindConnect and Open IoT OS. The IoT Edge Platforms such as LoopEdge and ClearBlade bring exclusive solutions to the table, targeting a range of hardware platforms like NXP, Dell, and Arduino, with trial and pay-as-you-go access models, respectively. This diverse array of platforms underscores the evolving intersection of EC, FL, and IIoT, offering tailored solutions.

5. Navigating challenges: A comprehensive examination of the problem statements and its solution techniques (approaches)

In this section, we critically evaluate the problem landscape, existing techniques, and methodologies used to address these challenges. This exploration highlights solutions proposed by different authors along with the examination of the diverse techniques presented; additionally we have presented the

Tables 4, 5 and 6 summarizing the "discussed research" of our work. The comprehensive summaries presented in Table 4, 5, and 6 provide an extensive overview of the state-of-the-art research in FL and edge computing within the IIoT domain. These tables serve as a foundational resource for understanding the diverse methodologies, models, and strategies employed to tackle critical challenges in this field. Table 4 focuses on privacy-preserving techniques, highlighting innovative methods such as multi-exit frameworks, secure aggregation, and differential privacy models. It has shown the balance between maintaining data privacy and achieving effective model convergence. This table also delineates the trade-offs between security and performance metrics such as accuracy, latency, and robustness. Table 5 delves into approaches for anomaly detection and fault diagnosis in IIoT networks. The summarized methodologies range from hybrid learning architectures to adaptive meta-learning techniques, showcasing their effectiveness in detecting system irregularities and minimizing downtime. By consolidating these works, the table emphasizes the significant role of FL in enabling resilient and adaptive IIoT systems. Table 6 addresses networking and resource management challenges, particularly focusing on communication overhead, bandwidth optimization, and data heterogeneity. It highlights the contributions like reinforcement learning-assisted offloading strategies and hierarchical frameworks that integrate edge and cloud resources.

Together, these tables provide a structured representation of the discussed research, a comparative understanding of existing techniques, identifying research gaps in federated learning and edge computing for IIoT applications.

5.1. Privacy-preserving techniques in federated learning

The preservation of data privacy has been an important aspect of FL training and modeling, especially when the Edge device is being used for the computation. Many such methods for privacy preservation (PP) have been employed to protect sensitive information throughout the FL training [15,20,22,24–32]. In addition, many works have proposed multi-objective techniques with different goals in mind such as global learning (GL) [33], solder paste printing (SPP) [34], fault detection (FD) [35] and training global models (TGM) [36] along with privacy preservation.

In [15], the primary focus is on PP while leveraging deep learning-based traffic predictors. The proposed approach, named Clustering-based Hierarchical and Two-step-optimized Federated Learning (CTFed), introduces a novel methodology. The local model has been optimized using PSO algorithm using two-step verification. A key attribute is its emphasis on training efficiency, ensuring accurate prediction performance, and maintaining robustness in the face of varying conditions.

Table 3
Listing some of the existing Edge-Cloud platforms, their key incorporation, and targeting hardware platforms.

Edge-Cloud Platforms	Key Incorporation	Target Hardware Platform
Google Cloud IoT	AndroidThings/Linux kernel	Cloud ML, Cloud Dataflow, BigQuery, Bigtable
MS Azure IoT	Linux/Windows	Azure IoT Edge, IoT Central
AWS IoT	Various (AWS Partner Network)	AWS IoT Analytics, Amazon FreeRTOS, IoT Device Management, IoT Core & device, AWS Greengrass
IBM Watson IoT	Linux, Windows	Watson IoT Cloud, Watson API
Bosch IoT Suite	Bosch IoT Gateway	Bosch IoT Gateway, Bosch IoT Remote Manager
Siemens Mindsphere	Siemens hardware	Open IoT OS, MindConnect
ClearBlade IoT Edge Platform	Various (NXP, Dell, HPE, etc.)	IoT Packet Manager, IoT Portal, EdgeLite,
LoopEdge	Litmus Automation	IoT Device Driver and Gateways, Connector

In [32], a hybrid architecture for Privacy-preserving Neural Network Training (PpNNT) was adopted to maximize resource utilization. This study emphasizes security, efficiency, and scalability, contributing to the development of PP solutions. The hybrid architecture achieves optimal resource utilization, aligning with the overarching goal of enhancing privacy and efficiency in IIoT environments. The challenge of learning models, and upholding data privacy is addressed in [33]. The proposed semi-decentralized federated edge learning (SD-FEEL) framework, coupled with edge aggregation optimization demonstrated improvement in accuracy. The DT-based Asynchronous SplitFed Learning is proposed in [69] to handle the data privacy, communication efficiency, and resource heterogeneity within e-healthcare systems. It demonstrates a focus on improving the efficacy of e-healthcare systems while considering resource variations and maintaining robust data privacy. In [47], data privacy and security concerns using adaptive federated multitask learning is addressed which uses a sparse sharing structure by employing an iterative pruning network. It leads to the generation of subnets for each task, contributing to an intricate yet accurate system that successfully achieves both accuracy and system scalability. The challenge of data heterogeneity in the IIoT context through their proposed Federated skewness Analytics and Client Selection mechanism (FedACS) is addressed in [65].

To preserve privacy, the data skewness quantification is in focus, and implementation demonstrates notable advantages, including optimized overheads, infrastructure reuse, and privacy preservation; making it a promising solution for handling data heterogeneity. In [24], the data privacy is being achieved using Multi-Exit-Federated edge learning (ME-FEEL) framework. It has provided the segmentation of deep models into several sub-models, each with different depths and output predictions. The framework adopts a greedy based exit selection strategy, enhancing the adaptability and efficiency of the learning process. The ME-FEEL is identified for its performance in achieving high accuracy. The data privacy concerns have been resolved through a multi-objective optimization problem [25]. The proposed solution involves a Deep Reinforcement Learning (DRL) algorithm based on deep deterministic policy gradient. It has designed to optimize multiple objectives simultaneously, including latency, energy consumption, and model accuracy. Additionally, the [27] focuses on data privacy uses Quantization-based Distributed Federated Learning (Q-DFL). It is designed to strike a balance among various factors, with a specific emphasis on system information trade-off. While this work neglects to address accuracy, it evaluates essential metrics such as the system's energy and flow consumption, as well as system time delay.

The [28] has centered the research on enhancing learning performance while meticulously preserving client privacy. It introduces a client selection method, termed Accuracy Client Selection (ACS), to refine the process of evaluating clients. This work has also neglected to measure the accuracy level of the system, however, it prioritizes metrics like accuracy and F1-score (Combined measurement of precision and recall) as key indicators of the learning system's performance. The synergies data filtering method named "Safe" has been introduced in [29], presents the collaborative power of Edge and cloud computing for the implementation of FL; emphasizes the safety and security of the distributed data. This study addresses the concern in distributed systems by proposing a PP approach that combines enhanced data security and trust. It provides a robust and reliable data handling mechanism along with accuracy and trustworthiness.

The critical issue of safeguarding private data, specifically for secure video offloading schemes, using multi-cell F-DRL is presented in [22]. The resource allocation and video frame resolution are achieved using a collaborative optimization problem. The twin delayed deep deterministic (TD3) policy gradient using a hierarchical reward function enhances the utilization of multi-cell horizontal FL. The research takes a comprehensive approach, considering factors such as utility, cost, security rate for a robust evaluation, energy consumption, and latency. In [30], the focus is on addressing challenges related to high-quality data, class imbalance, and poor diversity within the context of Industry 4.0. The proposed solution involves a decentralized architecture, employing multi-party sharing along with the privacy-enhancing scheme. In terms of learning, the research prioritizes data sharing through the implementation of the Findable, Accessible, Interoperable, and Reusable (FAIR) principles.

Differential privacy adds noise to the gradient updates, ensuring that individual data points cannot be reconstructed from the model. Techniques like secure aggregation allow the aggregation of model updates without exposing individual contributions by encrypting updates before transmission and decrypting them only after aggregation. Additionally, federated gradient masking and obfuscation can further anonymize data sources, thereby enhancing privacy. These methods are crucial for regulatory compliance in sectors where IIoT data confidentiality is mandated, such as manufacturing and healthcare.

5.2. Privacy leakage in IIoT

Addressing the critical concern of preventing privacy leakage due to the exposure of gradients, [38] introduces a DP FL approach. To effectively reduce the consumption of each global iteration privacy budget, the sparse vector technique is used. It achieves an application result, signifying improvement in terms of accuracy, practicability, and overall superiority. This research has address the privacy concerns in

Table 4
List of existing works addressing security within the framework of Edge-enabled federated learning in IIoT.

Ref.	PS	Model/ Architecture Description: N = Network,	Methodology A = Application, ML = Machine Learning	Performance N	A	ML	Metrics
[24]	Privacy preserving	ME-FEEL framework	Deep model, different depths and output prediction, greedy exit selection		A		Accuracy
[25]		Multi-objective optimization	DRL based deterministic policy gradient	N	A		Latency, EC, MA
[26]		Training ML models with IIoT devices	Robust aggregation rule, (AutoGM)		A		Robust, poisoning attacks
[27]		Q-DFL	Trade-off among the system information	N	A		EC, FC, delay
[28]		Client selection mechanism	ACS		A		Accuracy, F1-score
[29]		DF Edge-Cloud FL framework	Synergies in data filtering method		A		Accuracy, trustworthiness
[15]		CTFed	Particle Swarm Optimization (PSO) and a two-step approach for LMO		A	ML	Training efficiency, prediction, and robustness
[22]		Edge-Cloud integrated FL	Two-stage training mechanism		A	ML	Generalization ability, privacy protection
[20]		Hierarchical framework with FCT	Collaborative cloud–Edge–End structure and ML and EM for trustworthy localization		A	ML	Accuracy and convergence
[30]		Decentralized architecture	Multi-party technologies	N	A		Data sharing, FAIR
[31]		FL privacy-preserving collaborative ML	Tradeoff, data aggregation		A		Latency
[32]		PpNNT, hybrid architecture	Maximization resource utilization		A		Security, efficiency, and scalability
[33]		Semi-decentralized	Edge aggregation		A		Accuracy
[34]		Intelligent DSS across various networking devices	Decentralized, global machine intelligence model	N	A	ML	Wastes from trial production runs
[35]		Few-shot FD problem, generic framework,	Enhancement technique		A	ML	Convergence, accuracy
[36]		CSFL	MARL		A		Accuracy
[37]	Privacy leakage	DLF for training a global model	Joint Edge device selection Lyapunov optimization theory		A	ML	Cost efficiency, FL training performance
[38]		DP-FL	Sparse vector technique		A	ML	Accuracy, practicability and superiority
[39]		Fl based on data value evaluation	Data value evaluator designed			ML	Accuracy
[40]		FL architecture	Trade off solution		A	ML	Detection
[41]		FL, Autoencoder, Transformer, and Fourier mixing sublayer	Distributed ADA	N		ML	Communication costs, bandwidth consumption
[42]		Adaptive personalized federated meta-learning	OL rate adaptation, model-agnostic meta-learning			ML	Accuracy, loss function
[43]		Fed-SCR, Fed-GMA	Semisupervised generative network		A	ML	Classification accuracy
[44]		Distributed AD system	Physical layer feature, CFO	N	A		Overhead, robustness
[45]	Attack and privacy threats	FL-based detection protocol	Mitigation intelligence		A		Accuracy, eliminates mitigation time, cost
[46]		Reliable AD strategy using FL	Local model trained by DRL	N	A		Throughput, latency, and AD accuracy
[47]		AFL	Sparse sharing structure, iterative pruning network and generating subnets	N	A	ML	Accuracy and system scalability
[48]		Federated semisupervised learning scheme	AE, FL, supervised neural network by FCN		A	ML	Detects attacks with high classification
[49]		FL-CTIF for securing IIoT environments	Comprehensive cyberattack dataset with updated FS using IF		A		Accuracy, F1-score, FPR
[50]		FL based Deep Intrusion Detection	Collaborative model, Paillier-based encryption, Neural Network for ID model		A		Effectiveness
[51]		Adaptive differential privacy-based FL, RL	Network intrusion		A	ML	Efficacy, privacy, accuracy
[52]		FDA[3]	Cloud-based architecture		A		Resist
[53]		Fed-IIoT	A3GAN, GAN-based countermeasure		A		Robustness, privacy
[54]		Targeted ransomware detection model	Peer-to-Peer Federated Learning (AP2PFL) and Deep Learning (DL)		A		Detecting
[49]		FL-CTIF for securing IIoT environments	Comprehensive cyberattack dataset with updated FS using IF		A		Accuracy, F1-score, FPR

the context of traffic prediction, offering a comprehensive solution which excels in training efficiency and predictive accuracy. The DP techniques is utilized to add noise or perturbation to the model updates, ensuring that individual data samples cannot be inferred. Secure aggregation methods, such as secure multi-party computation (MPC) or homomorphic encryption, enable ED to aggregate model updates without directly accessing each other's data, maintaining privacy while still generating an accurate global model. In response to concerns about the leakage of private information, [39] proposes a FL framework centered around data value evaluation. It involves the design of a evaluator for data value situated in the cloud, responsible for assessing the significance of data contributed by each client; achieves high accuracy in its predictions. Differential privacy (DP) and secure aggregation (SA) are key techniques used to address privacy concerns in FL. DP introduces noise to the model updates, thereby preventing reverse engineering of individual data samples from aggregated updates [38]. Secure aggregation, on the other hand, leverages cryptographic techniques to allow aggregation of updates without revealing individual model parameters, often using protocols like homomorphic encryption to aggregate updates securely [26].

Table 5
List of existing works addressing security within the framework of Edge-enabled federated learning in IIoT.

Ref.	PS	Model/ Architecture Description: N = Network,	Methodology A = Application, ML = Machine Learning	Performance N	A	ML	Metrics
[55]	Fault detection	ABS using MDP	Optimize resource scheduling and IL offloading task	N	A		EC and latency
[56]		Duplet classifier	Mixed fault classification		A		Accuracy, robustness
[21]		HLFI	Hierarchical federated learning strategy and dynamic elimination learning strategy		A		F1-score
[57]		Fed-TSN controller framework					
[58]		Data-sharing	Convolutional denoising AE and FSML		A		Accuracy
[59]	Model train	CM CC	MDP			ML	QoS
[60]		DFL	Consensus in DFL as a fastest mixing MDP		A		Convergence rate
[61]		FL (PFL) framework	Robin Hood mechanism			ML	Training loss
[62]		FL framework	Min–max mixed integer NLP based on the Lyapunov technique		A	ML	EC, and computation frequency, convergence time

5.3. Anomaly detection (AD) using FL

In [41], the AD challenge is addressed by detecting irregularities in time-series of data through a hybrid design that incorporates FL, Autoencoder, Transformer, and a Fourier mixing sublayer. The proposed architecture emphasis on low communication costs and reduced bandwidth consumption. To fingerprint the devices, Carrier Frequency Offset (CFO) method is used from physical layer in [44]. It enables the system to effectively detect anomalous deviations, enhancing the overall robustness against cyber attacks with minimal overhead. In the domain of trustworthy AD within smart Grids, [43] proposed a privacy-preserving federated semisupervised class-rebalanced (Fed-SCR) framework, incorporating federated geometric median aggregation (Fed-GMA). The minority samples is generated for model relationships between labeled and unlabeled data; the quality is being enhance using semisupervised generative network. It has used temporal convolutions as leveraged to improve representation power during training, leading to robust classification accuracy of AD for smart grids. The [46] presents a reliable AD strategy designed for the IIoT, leveraging FL. The proposed approach focuses on addressing security and privacy threats through the implementation of a universal AD model. The local models are trained using the deep reinforcement learning (DRL) algorithm, resulting in a system characterized by high throughput, low latency, and exceptional accuracy in AD along with particular attention to PP.

5.4. Malicious attack (intrusion detection (ID) and prevention) using FL

In [49], FL-based Cyber Threat Intelligence Framework (FL-CTIF) for enhancing the security of IIoT presented. The framework leverages virtual representation and ID systems. This work advocates for a comprehensive cyber attack dataset and apply information fusion on updated feature selection. The evaluation metrics, including accuracy, reduction in the false-positive rate, F1-score, and underline the effectiveness of the FL-CTIF framework in fortifying IIoT security.

An FL-based detection protocol tailored for the IIoT is introduced in [45], where it addresses malicious code attacks by employing distributed and collaborative mitigation intelligence. With a strong emphasis on accuracy, the protocol efficiently eliminates mitigation time and reduces associated costs. The approach demonstrates high performance, acceptable accuracy and the overall efficacy of the mitigation process. In [48], it tackles the challenge of attack detection in the IIoT domain by employing a federated semisupervised learning scheme. The approach utilizes autoencoders (AE) to extract low-dimensional features from unlabeled local/private data. The AE is integrated into the FL framework and enhanced with supervised neural networks by incorporating fully connected layers (FCN) to the global encoder. The evaluation shown the effectiveness of system in detecting attacks with high classification performance. The [50] presents an ID framework for Industry 4.0 that incorporates FL as collaborative model to detect threats effectively, addressing the limitations of individual devices. Communication between ED is secured using Paillier-based encryption. The ID model employs a combination of Multi-Layer Perceptron, Long Short Term Memory, and Neural Network.

The [51] proposes a data-driven cyber-defense mechanism for the electric vehicle (EV) charging ecosystem. Their adaptive differential privacy-based FL framework incorporates a RL-based intelligent privacy allocation and provided the collaborative network ID system. In [26], the authors address the training FL models across multiple ED while ensuring the robustness of the models against various challenges, including potential poisoning attacks. It propose a novel aggregation rule named auto-weighted geometric median (AutoGM), emphasizing its robustness.

5.5. Security vulnerabilities in edge devices and FL mitigation strategies

Edge devices in IIoT environments are susceptible to unique vulnerabilities, including physical tampering, side-channel attacks, and targeted malware, which pose risks to data integrity and model reliability. In case of FL, it mitigates some of these vulnerabilities by avoiding centralized data storage, which reduces the attack surface. Furthermore, secure aggregation protocols, such as homomorphic encryption and secure multi-party computation, are applied to protect model updates during transmission. By preserving data locality and employing cryptographic methods, FL can provide robust security for edge-deployed IIoT systems, even in cases where physical security cannot be guaranteed.

5.5.1. Recent advancements in privacy-preserving techniques for FL in IIoT

Adaptive Federated Averaging (AdaFed) has emerged as a pivotal technique in addressing the non-IID nature of data in IIoT. Unlike traditional federated averaging, AdaFed dynamically adjusts the contribution of each client's local model update based on factors such as data distribution, quality, and device reliability [7]. This adaptive mechanism not only enhances model convergence but also ensures that privacy-preserving objectives are met by minimizing the exposure of sensitive gradients from unreliable devices.

Another significant advancement is the integration of differential privacy (DP) models in FL, which introduces noise to local updates before aggregation, ensuring that individual data points cannot be inferred from the shared parameters [12]. Techniques like zero-concentrated differential privacy (zCDP) and Rényi differential privacy

Table 6
List of existing works addressing networking parameters within the framework of Edge-enabled federated learning in IIoT.

Ref.	Problem statements Description: N = Network,	Model/ Architecture A = Application,	Methodology ML = Machine Learning	Performance N	A	ML	Metrics
[63]	Data Management	FL algorithm assisted by DRL	Equipment selection process	N		ML	Aggregation rate and reduce communication costs, accuracy
[64]	Heterogeneous data management	Federated group synchronization	Client selection strategy, collaborative training protocol	N		ML	accuracy, loss, and faster convergence
[65]	Heterogeneous data management	FedACS	Quantify the data skewness	N	A	ML	Privacy preservation, optimized overheads
[66]	Offloading	Q-Learning-Aided Offloading	FL loss FM with LC	N			Accuracy
[67]	Optimal resource allocation	FedBarg, a game-theoretic scheme	Non-convex optimization	N			Delay, revenue
[14]	Efficient management and QoS	D2D communication-aided DT Edge network	DT-DFL	N			Network performance
[68]	Mobile crowdsensing	Incentive-based FL	DT-driven Stackelberg-based inter-client	N	A	ML	Accuracy, fairness, and data compatibility
[69]	Heterogeneity, communication, and data privacy	DT-ASFL	SplitFed Learning to introduce communication efficiency		A	ML	Efficacy
[70]	Semantic Communication (SC)	FL and A3C	transfer learning	N	A	ML	Accuracy
[71]	Data transmission energy consumption	Two-stage iterative optimization framework, ERM	Jointly minimize the global transmission energy's	N		ML	TEC, training energy
[72]	Confidentiality, class distribution and degradation	FED GS	GBP-CS, FL training, compound-step synchronization protocol	N	A	ML	Robustness, accuracy, efficiency, reduces training
[73]	Network optimization, service prediction	BrainIoT	Industrial knowledge graph-based relation mining, FL service prediction, and GOR reservation	N	A		Service prediction, accuracy, and QoS
[74]	Dedicated scheduling	Contention-based design for uplink transmissions	URLLC and FL traffic	N			Re-transmissions rate
[75]	Communications overhead and privacy vulnerability	DT architecture	Decentralized FL and SL to construct the DT models	N	A		reliability and EC
[76]	Timeliness of the MEC systems, and computation tasks	Multimodal deep RL, H-MAAC	Online multi-agent cooperation, joint MEC collaboration		A	ML	Average system age, stability of training process
[77]	Thermal comfort prediction, privacy-preserving	Fed-NN	branch selection protocol, communication overhead		A	ML	Robustness, accuracy, and stability
[78]	Supply chain management	Harmonized Supply Management Scheme (HSMS)			A		Delivery rate and product availability
[79]	Inaccessibility of private data, flows disruptions, machines' breakdowns, quality crisis	Localization of mobile resources, tasks rescheduling optimization	FDL as distributed learning technique for prediction model	N	A	ML	Prediction accuracy
[80]	Intelligent decisions, data annotation	Generic framework for assembly and disassembly lines	WeCollab, self-supervised learning		A	ML	Test accuracy
[81]	Collaborative learning process for communication	DetFed	Multi-objective stochastic optimization problem	N		ML	FL Convergence and learning accuracy
[82]	Network-as-service	Deep RL scheme for network management and resource allocation	Multi-agent self-model, optimal action decision	N		ML	Efficiency
[83]	Resource Allocation	RaFed	Optimization problem		A	ML	Latency
[84]	Feature extraction	J-HOOI	FTD		A	ML	Accuracy
[85]	Communication overhead, DT	DITENs	Optimization problem	N		ML	Efficiency
[86]	Resource allocation	DRL-FL	Optimization	N			Efficiency
[87]	Trusted-based aggregation	DRL	Aggregation frequency of FL on Lyapunov dynamic deficit queue and DRL		A	ML	Learning accuracy, convergence, energy saving
[88]	Aerial Remote Sensing	FRL	UAV clusters	N		ML	Centralized
[89]	Predictive maintenance	Sampling and data reduction techniques		N	A		Learning accuracy, convergence

(RDP) have been adapted for FL scenarios, offering improved privacy guarantees while balancing the trade-off between model accuracy and privacy loss [38,90,91]. These methods are particularly relevant in IIoT, where devices operate under stringent computational and communication constraints.

Furthermore, recent studies have proposed hybrid privacy-preserving frameworks that combine encryption-based secure aggregation with DP models to address adversarial threats comprehensively. For instance, federated learning frameworks employing homomorphic encryption alongside DP mechanisms ensure robust protection against data inference attacks without introducing significant computational overhead [92]. Techniques such as hierarchical FL, where intermediate aggregators secure partial updates before final aggregation, reduce communication bottlenecks while maintaining strong privacy guarantees. Additionally, trust-based client selection algorithms further mitigate risks by prioritizing devices with reliable data and communication capabilities [92].

5.6. Network communications techniques using FL over edge

Effective network communication is critical for enabling FL over edge in IIoT. This subsection delves into techniques aimed at optimizing communication efficiency, addressing bandwidth constraints, and ensuring robust data exchange in dynamic and distributed edge environments. The [14] proposed a digital twin (DT) Edge network using device-to-device communication to achieve efficient network management and QoS. The proposed framework leverages digital twin technology and integrates FRL to enhance the network performance to

elevating the quality of service (QoS) and overall efficiency. In [73], they have addresses the challenges of network optimization, aiming to increase the service prediction accuracy. It introduces a service provisioning scheme using brain-like thinking leveraging FL named as "BrainIoT". The scheme incorporates three algorithms named as globally optimized resource reservation, FL-based service prediction, and industrial knowledge graph-based relation mining. The work demonstrate that it has significantly improves the accuracy of service prediction, ensuring a higher QoS. In [82], they presented a network-as-a-service solution for the IIoT leverages a deep reinforcement learning (DRL) scheme to offer federated and dynamic network management along with resource allocation. The proposed deep Q-learning-based process adjusts slices for throughput and spectral efficiency dynamically. Additionally, DRL is introduced to enable multiagent self-model learning, facilitating optimal action decisions demonstrates improved efficiency.

The 5G NR IIoT faces contention on large-scale often faces challenges in scheduling, which handled using dedicated scheduling mechanism presented in [74]. The network simulations 3GPP-compliant is used for up-to-date analysis of network parameters. Their study extends to application-level performance assessment, specifically addressing simultaneous Ultra-Reliable Low Latency Communications (URLLC) and FL traffic. The results underscore the significance of dedicated scheduling in reducing re-transmissions of lost or collided transmissions, thereby optimizing network efficiency and reliability. The [81] considers the ultra-reliable application and introduce a collaborative learning process. Their proposed deterministic FL framework, named "DetFed", addresses a multi-objective stochastic optimization problem. This framework aims to simultaneously maximize the scheduling success ratio and learning accuracy while ensuring deterministic requirements related to delay, jitter, low-latency and packet loss. The approach exhibits improved FL convergence and enhanced learning accuracy.

The [68] presents a novel approach to mobile crowdsensing using an incentive-based FL scheme. The method incorporating a combination of Stackelberg-based inter-client incentive along with contract-based reputation. This work is driven by DT technology, offering a sophisticated architecture for industrial mobile crowdsensing. The key focuses of their contribution include improving accuracy, ensuring incentive fairness, and enhancing data compatibility. The Semantic Communication (SC) approach integrates Asynchronous Advantage Actor–Critic (A3C) with FL techniques [70]. In system control transmission energy consumption for long-term data by focusing on sampled data sampling, and freshness at the controller is addressed in [71]. Its two-stage iterative optimization framework, employs an entropic risk measure (ERM). The primary goal is to jointly minimize the global transmission energy consumption, demonstrating a nuanced understanding of energy consumption and training energy through the utilization of ML techniques.

In [20], a hierarchical framework leveraging federated control theory is proposed. The approach adopts a collaborative cloud–Edge–End architecture and integrates ML localization, specifically utilizing the expectation maximization (EM) method for trustworthy localization. The research emphasizes accuracy and convergence as key evaluation metrics for the proposed framework. The [72] proposed FED GS, a hierarchical cloud–Edge–end FL framework. It contributes to the challenge of maintaining the confidentiality of industrial data. In addition, it addresses the class distribution issues that could degrade the performance of FL in the context of IIoT and 5G-empowered Industry 4.0. The novelty lies in the application of a gradient-based binary permutation algorithm (GBP-CS) for selecting the subset of devices within each factory. This approach aids in enhancing the participation in FL training through super homogeneous nodes. Additionally, a synchronization protocol using compound-step is proposed, which evaluated through non-availability of device attacks, shown improvements in robustness, accuracy, efficiency, and a reduction in training complexities. The [36] focuses on data privacy concerns while training global models using multi-agent reinforcement learning (MARL) and Cross-Silo FL (CSFL) approach.

5.7. Networking protocols and topologies in FL for IIoT

Certain networking protocols are particularly suited for FL in industrial settings due to their support for low-latency and high-reliability communication. Protocols such as Message Queuing Telemetry Transport (MQTT) and Constrained Application Protocol (CoAP) are commonly used in IIoT due to their lightweight nature and efficiency over constrained networks [72]. Additionally, Time-Sensitive Networking (TSN) standards are increasingly adopted in industrial FL to ensure deterministic data transmission, which is critical for time-sensitive IIoT applications. Mesh topologies are also preferred as they allow direct communication between devices, reducing reliance on central hubs and enhancing fault tolerance within FL workflows.

5.8. Training personalized models for IIoT

In Industrial Internet of Things (IIoT) environments, the diversity of devices and data distributions necessitates personalized model training approaches [42,59–62]. This subsection explores strategies and methodologies for adapting personalized federated learning (FL) models to heterogeneous edge devices. In [59], the local DRL model training limitation was addressed in the IIoT context. The cross-modal collaborative communication utilizing a Markov decision process (MDP) to enhance the QoS was presented. The [60] used decentralized federated learning (DFL) to perform training on a shared model. The consensus achievement in DFL used for treating it as a fastest mixing Markov chain problem to achieve faster convergence rate. The [61] presents a study on training personalized models in the context of the IIoT. It proposes a PFL framework that incorporates game-theoretic insights and transferable utility, and specifically employs the Robin Hood mechanism, indicates a significant contribution with a focus on training loss. In [62], large-scale DNNs challenge of local model training has been addressed in the context of the IIoT. The FL framework aimed to minimize the training delay by adopting the communication–computation efficiency. The optimization is performed under constraints such as device-, memory usage, energy consumption, and specific participation rate. This work focus on optimizing DNN partitioning, transmit power, channel assignment, convergence time, and computation frequency. In [42], based on online learning rate adaptation method an adaptive personalized federated meta-learning framework proposed which employs a dynamically adjusting hyperparameters during the training process. The mechanism model meta-learning contributed to enhanced accuracy and optimized loss functions for industrial equipment. In [85], a framework named "DITENs" was proposed to minimize the communication overhead of FL. It focused on optimizing the communication process, acknowledging the importance of reducing communication costs in FL. The work is evaluated in terms of accuracy, but is significant for improving efficiency. This emphasis on efficiency implies that DITENs could play a crucial role in scenarios where communication efficiency is a critical factor, such as large-scale FL environments with distributed and interconnected nodes.

The [76] was focused on addressing concerns related to the timeliness of MEC, freshness of data, and computation tasks. The authors proposed a multimodal DRL framework named Heterogeneous Multiagent Actor–Critic (H-MAAC) which is an online algorithm, incorporating Edge FL into multiagent cooperation. The approach not only guarantees theoretical advantages in learning convergence but also contributes to improving the average system age and the stability of the training process. In [64], they addressed the challenge of handling heterogeneous data through a federated group synchronization framework. The proposed model is focused mostly on shared training and client selection. The application of this framework resulted in improved accuracy, lower loss, and faster convergence, showcasing its effectiveness in handling diverse datasets. Training global models using distributed learning and prevention of privacy leakage issue was addressed in [37]. The bandwidth allocation and joint ED selection is used for Lyapunov optimization theory for efficient utilization of virtual energy deficit queue; able to provide cost efficiency and performance improvement in FL training.

5.8.1. Personalization, incentive mechanisms, and fairness in FL for IIoT

To ensure equitable and efficient participation in FL, personalization, incentive mechanisms, and fairness have become essential components in IIoT deployments:

- **Personalization:** Due to the heterogeneity of IIoT devices, personalized FL models are necessary to cater to varying device capabilities and data distributions. Personalized FL algorithms, such as adaptive model tuning [32], allow each device to adjust the global model to better fit its unique operational context.
- **Incentive Mechanisms:** Devices in IIoT require motivation to participate in the training process, especially when resource consumption is involved. Approaches like federated reinforcement learning-based incentives [87] have been applied to encourage device collaboration by offering rewards based on data quality and contribution.
- **Fairness:** Ensuring that each participant's data is represented fairly; techniques that adjust for class imbalance and data heterogeneity, such as FedAvg and FedProx, have been adopted to improve fairness, especially in applications like predictive maintenance and fault detection [15].

5.9. Resource management

The resource management has critical presence due to wide range of resources such as network bandwidth, computation allocation, and storage which directly affect the overall system performance in term of accuracy, reliability, system efficiency, robustness [84]. In this context, the [83] presented "RaFed", a framework focusing on resource allocation and formulated as an optimization problem, leveraging ML techniques, with an emphasis on achieving low latency. The [84] introduced "J-HOOI" a framework designed for feature extraction, using "Feature Transformation and Distillation" (FTD). The framework was evaluated in terms of accuracy and shown improvement for tasks where precise feature extraction is crucial for achieving high-performance results.

The optimal resource allocation in IIoT through their proposed scheme, FedBarg is considered in [67]. It employed utility theory-based joint coalition formation game theory. The non-convex optimization problem, emphasizing scheme's potential to resource allocation. It aimed at minimizing delay and maximizing revenue. In [66], it presents a data offloading scheme using Q-Learning-Aided Offloading Strategy. It address the FL loss function minimization along with latency constraints. It showcases a non-negligible advancement in accuracy, significance in optimizing offloading processes.

A DRL using FL approach named DRL-FL, with a specific focus on addressing challenges related to resource allocation is proposed in [86]. The overall efficiency of FL systems was improved using resource allocation optimization. In addition, [87] also introduced a DRL using FL approach, termed DRL-FL, that centers around trusted-based aggregation mechanisms. In this work, the aggregation frequency is dynamically adjusted using Lyapunov dynamic deficit queue and DRL techniques. The proposed work is evaluated based on energy efficiency, convergence, and learning accuracy shown improvement. The integration of trust-based mechanisms suggests that DRL-FL addresses challenges related to secure and efficient aggregation in FL setups, making it particularly relevant in scenarios where maintaining the integrity of the aggregated model is crucial.

The [88] presented an approach known as FRL, designed for Aerial Remote Sensing, specifically focusing on UAV clusters. Unlike many FL frameworks that operate in a decentralized manner, FRL takes a centralized approach. The utilization of UAV clusters in a centralized FL model signifies a departure from the typical decentralized setup. It might be attributed to specific requirements or constraints in the context of aerial remote sensing.

Centralized resource allocation relies on a central server to manage and distribute resources across edge devices. This approach simplifies management and ensures uniformity in decision-making. Conversely, decentralized approaches distribute the decision-making process among edge devices, leveraging local knowledge and peer-to-peer communication. Centralized approaches are often constrained by the computational capacity of the central server, resulting in potential bottlenecks in low-bandwidth or high-latency networks. On the other hand, decentralized strategies excel in scenarios where edge devices have varying levels of computational capabilities, as they enable localized optimization and reduce dependency on a single point of control. Decentralized approaches inherently provide better fault tolerance and scalability, as resource allocation decisions are distributed and less reliant on a single node. However, they require robust consensus algorithms to ensure consistency, which can add computational overhead and complexity. In contrast, centralized approaches, while easier to implement, are susceptible to single points of failure and may struggle to scale efficiently in large networks [83]. For instance, in a smart manufacturing setup, centralized allocation may be preferred for tasks requiring high precision and strict timing, where the central server's computational capacity is sufficient to handle the workload. Decentralized strategies, on the other hand, are better suited for geographically dispersed systems, such as agricultural IoT networks, where edge devices operate independently and face diverse environmental conditions. Table 7 has provides a comparative summary of centralized and decentralized strategies.

5.9.1. Trade-offs

The trade-offs between centralized and decentralized approaches can be summarized as follows:

- **Centralized Approaches:** Simplified management and decision-making but limited scalability and resilience to failures.
- **Decentralized Approaches:** Higher fault tolerance and adaptability but require complex algorithms and may introduce computational overhead.

5.10. Predictive maintenance, fault diagnosis (FD) and supply chain optimization

The predictive maintenance, employing sampling and data reduction techniques is proposed in [89]. It implied an exploration of efficient methods for handling and analyzing data in the domain of predictive maintenance. In smart manufacturing, the importance of intelligent decision support among different networking devices is addressed in [34] for solder paste printing (SPP). It involves a global machine intelligence model aggregated in a decentralized and PP manner for the novelty, applicability, and employment of ML. This approach contributes to minimizing wastes resulting from trial production runs, demonstrating its relevance and potential impact on enhancing efficiency in the manufacturing process. In the domain of mechanical fault diagnosis (MFD) in the IIoT, [35] addresses the challenge of few-shot FD problems through the development of a generic framework. It has shown faster convergence and higher accuracy, contributing significantly to the field of MFD while prioritizing privacy preservation.

A novel approach for failure prediction utilizing homophily learning-based federated intelligence (HLFI) was presented in [21]. Their proposed method incorporates a dynamic elimination learning and hierarchical FL strategy to achieve high F1-score in failure prediction. In [57], failure recovery in the IIoT has been address using FL based Time-Sensitive Networking (Fed-TSN) controller. The machinery fault diagnosis in the IIoT focused on data-sharing techniques and utilizing a feature-space metric learning along with convolutional denoising autoencoder is presented in [58] able to achieve high level of accuracy in FD. The FD in rotating machinery within the IIoT context is presented in [56] uses a duplet classifier. An innovative approach addresses mixed fault classification tasks by utilizing parallel networks achieving high

Table 7
Comparison of centralized and decentralized resource allocation strategies.

Aspect	Centralized approach	Decentralized approach
Management	Simplified	Distributed
Scalability	Limited	High
Fault Tolerance	Low	High
Overhead	Low	Moderate to High
Use Cases	High-precision tasks, low device diversity	Distributed systems, high device diversity

accuracy and robustness in FD.

In [55], they addressed the inability to use equipment as resources thoroughly in Industry 4.0. It propose an ambient backscatter system utilizing a Markov decision process. The focus is on intelligent algorithms to optimize a resource scheduling scheme for offloading tasks to reduce system energy consumption and latency. The [77] addressed thermal comfort prediction in IIoT settings while prioritizing PP. The proposed a FL-based neural network algorithm named Fed-NN. It uses a branch selection protocol to handle the issue of communication overhead. It is able to achieve robustness, accuracy, and stability in thermal comfort prediction. The [79] addressed the challenge of the inaccessibility of private data, flows disruptions, machines breakdowns, and quality crises. It propose a monitoring tool designed to detect system disruptions associated with the localization of mobile resources. The solution involves an optimization problem for tasks rescheduling, using FDL — a distributed learning technique. The approach enables the construction of a prediction model for resource locations, showcasing its effectiveness in achieving high prediction accuracy.

To address the supply chain management challenges within the IIoT a Harmonized Supply Management Scheme (HSMS) is proposed which aims to enhance the delivery rate and product availability [78]. It has enabled the intelligent decisions and data annotation for industrial robots involved in assembly and disassembly lines. Their proposed framework, named WeCollab, is designed to be generic and adaptable to various image data types generated by industrial robots. It leverages self-supervised learning methods to achieve its objectives, showcasing promising results with high test accuracy.

6. Challenges, observation and open issues

In the domain IIoT, when the combination of FL and EC comes together to solve specific problems; it presents several challenges, spanning a spectrum of domains, including resource allocation, communication overhead, and model optimization. Optimizing resource allocation in an FL environment is inherently complex challenge. The heterogeneous nature of devices and varying network conditions pose hurdles in achieving efficient resource utilization. Additionally, dealing with dynamic and unpredictable changes in network conditions and device availability affects the effectiveness of resource allocation strategies. Mitigating communication overhead in a dynamic industrial network, where frequent changes in topology and device connectivity occur [85] also hinders the resource allocation. Additionally, balancing the trade-off between resource efficiency and model accuracy, as optimizing for one may negatively impact the other. The dynamic industrial network, where the topology have frequent modifications can also present potential security and privacy concerns associated with resource allocation decisions in a FL. Considering the impact of communication delays on the overall performance of the industrial network and proposed strategies has tried to minimize these delays [85]. Additionally, the issue of scalability, especially as the number of devices participation FL increases is also address but need further attention.

Addressing interoperability challenges in the extracted features to enhance the understanding of the shared learning or GL presented in joint hierarchical optimization approach [84] have concern over data sharing. The amount and nature of shared data can significantly impact the training time, training quality, convergence, delay and decision making. Many have suggested different kind of shared training such as vertical [22], horizontal [15,20,21], and hybrid [72]; though they have avoid the training time and convergence details which need further attention [93–95]. In IIoT environments, edge devices often exhibit significant variations in data distributions due to differences in operating conditions, sensor configurations, and machinery types. This distributional discrepancy can hinder the effectiveness of GL approaches, where a central server aggregates updates from all participating devices [75]. Edge devices in IIoT networks typically have constrained communication bandwidth and intermittent connectivity. Transmitting model updates to a central server for global aggregation can incur significant communication overhead, leading to latency and efficiency issues. The Edge environments in IIoT are susceptible to network failures, device malfunctions [21], and adversarial attacks [52,53], which can disrupt FL processes and compromise model integrity. Ensuring fault tolerance and robustness requires resilient aggregation protocols, Byzantine fault-tolerant algorithms, and anomaly detection mechanisms to detect and mitigate malicious behavior or system failures [54].

Another challenge is to develop RL algorithms capable of effectively optimizing resource allocation [86,87]. The challenge of non-convex optimization in resource allocation, considering the complex and dynamic nature of federated environments. Ensuring the stability and convergence of the proposed DRL algorithms during the resource allocation process. Designing RL strategies that prioritize trusted aggregation nodes while considering the dynamic nature of trust levels in a federated setting. In addition, computational complexity of the reinforcement learning algorithms and ensuring their feasibility in real-world settings. Investigating the impact of the aggregation frequency on learning accuracy, convergence, and energy efficiency, and optimizing these trade-offs. Developing efficient mechanisms for model aggregation in a decentralized setting, where UAVs act as both learners and contributors [88].

6.1. Communication bottlenecks and their impact on model convergence in heterogeneous environments

Communication bottlenecks are a critical challenge especially in heterogeneous environments where ED have varying computational, network, and energy capabilities. Bottlenecks arise primarily due to limited bandwidth or high latency between edge devices and the central server, which significantly impacts model convergence. In such scenarios, delayed updates from slower devices, often termed stragglers, hinder the global model's ability to converge efficiently. This delay leads to gradient staleness, where the outdated gradients uploaded by lagging devices can result in suboptimal convergence or delay in the optimization process. Moreover, devices with poor communication capabilities might frequently drop out of training, reducing the diversity of data representation and adversely affecting the model's robustness and fairness [84].

Scalability is particularly challenging in IIoT systems with diverse hardware configurations and data distributions. For instance, non-IID data across devices amplifies the adverse effects of communication bottlenecks, as updates from underrepresented devices carry disproportionate importance in achieving balanced model accuracy [96–98]. Furthermore, high-performing devices often remain idle while waiting for slower devices, resulting in resource underutilization and inefficiencies in the learning process [81].

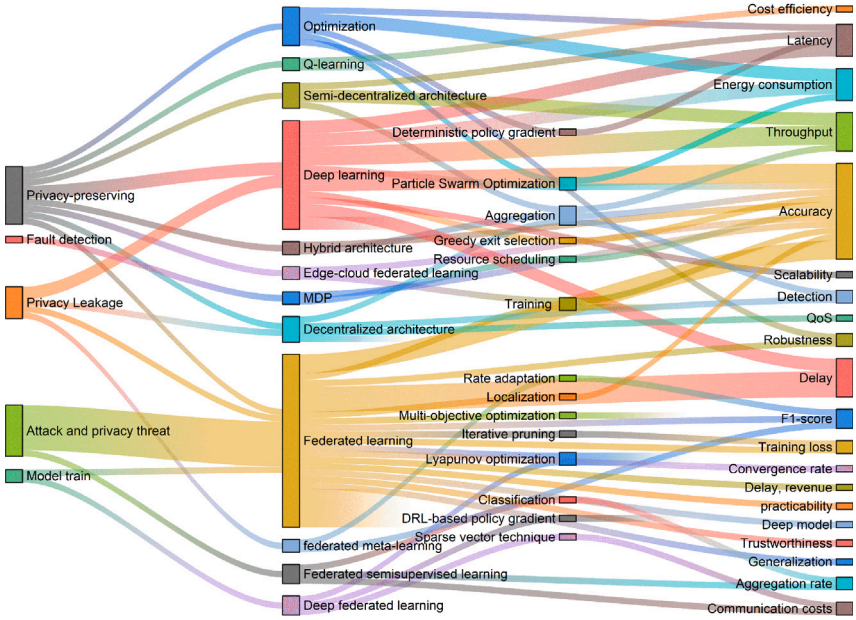


Fig. 4. KPI analysis based on the observation finding in of Tables 4 and 5.

Several strategies have been proposed to mitigate these issues; techniques such as asynchronous FL can reduce the reliance on synchronous updates by allowing faster devices to proceed independently, though additional mechanisms are needed to handle potential gradient staleness [25]. Hierarchical FL architectures, where intermediate aggregators handle local updates, alleviate communication loads on the central server and improve scalability in large-scale deployments [30]. Gradient compression and sparsification methods, which selectively reduce the size of updates, further minimize communication overhead without sacrificing model accuracy [96]. Additionally, prioritizing updates from devices with better communication capabilities or higher-quality data through weighted aggregation schemes has been shown to enhance the global model's reliability [12].

Addressing communication bottlenecks in FL requires a nuanced approach that balances efficiency with accuracy, particularly in heterogeneous IIoT environments. Integrating adaptive communication protocols, advanced aggregation methods, and hierarchical models will be key to achieving scalable and robust FL deployments.

6.2. Trade-offs between security and performance

While implementing security measures such as homomorphic encryption and secure multi-party computation improves data privacy in FL, these methods introduce latency and computational overhead. Homomorphic encryption, for example, requires additional computation during both encryption and decryption stages, leading to increased latency in environments where real-time decision-making is essential [8,32]. Similarly, secure aggregation methods, though essential for maintaining data integrity, demand higher communication bandwidth, which can limit scalability. Balancing these security requirements with performance metrics is crucial in IIoT applications, where strict latency constraints must be met.

The specific challenges identified through a our work require careful consideration and strategic interventions in the pursuit of advancing these cutting-edge technologies [21,52–54,75,96–98]

1. Optimizing resource allocation in a federated learning environment is inherently complex. The heterogeneous nature of devices and varying network conditions poses a challenge in achieving efficient resource utilization.
2. Dealing with dynamic and unpredictable changes in network conditions and device availability, which may impact the effectiveness of resource allocation strategies. Additionally, balancing the trade-off between resource efficiency and model accuracy, as optimizing for one may negatively impact the other.
3. Mitigating communication overhead in a dynamic industrial network, where frequent changes in topology and device connectivity occur [85].
4. Considering the impact of communication delays on the overall performance of the industrial network and proposing strategies to minimize these delays. Additionally, The issue of scalability, especially as the number of devices participating in federated learning increases.
5. Developing mechanisms to handle potential security and privacy concerns associated with resource allocation decisions in a federated setting.
6. Addressing interoperability challenges in the extracted features to enhance the understanding of the learned representations in joint hierarchical optimization approach [84].
7. Developing reinforcement learning algorithms capable of effectively optimizing resource allocation [86,87]. The challenge of non-convex optimization in resource allocation, considering the complex and dynamic nature of federated environments.
8. Ensuring the stability and convergence of the proposed Deep Reinforcement Learning algorithms during the resource allocation process. Designing reinforcement learning strategies that prioritize trusted aggregation nodes while considering the dynamic nature of trust levels in a federated setting. In addition computational complexity of the reinforcement learning algorithms and ensuring their feasibility in real-world settings
9. Investigating the impact of the aggregation frequency on learning accuracy, convergence, and energy efficiency, and optimizing these trade-offs. Developing efficient mechanisms for model aggregation in a decentralized setting, where UAVs act as both learners and contributors [88].

10. Scalability and communication overhead present major bottlenecks in deploying FL for IIoT applications. As the number of devices in IIoT environments grows, maintaining low latency and high accuracy becomes challenging due to limited communication bandwidth and the computational constraints of edge devices. Current approaches focus on model compression, efficient parameter updates, and optimized communication protocols to mitigate these issues; however, further research is needed to enhance scalability without compromising the system's performance and responsiveness.

To mitigate above mentioned challenge, researchers need to explore techniques for efficient model compression, quantization, and differential privacy-preserving aggregation to reduce the communication footprint while preserving model accuracy [96–98]. In addition, this challenge requires advanced techniques for distributed robust optimization or domain adaptation, ensuring that the central model can generalize well across diverse edge environments.

6.3. Observations using key performance indicator (KPI) analysis

The diagrams shown in Figs. 4–5 effectively highlight the interplay between problem statements, methodologies, and key performance metrics within the framework of Edge-enabled federated learning for industrial IoT (IIoT). Figs. 4 and 5 provide a comprehensive visualization of key insights derived from Tables 4, 5, and 6.

Fig. 4 delves into security-oriented applications within IIoT, spotlighting challenges like fault detection, privacy leakage, and attack mitigation. Methodologies such as decentralized architectures, edge-cloud federated learning frameworks, and deep learning-driven optimization approaches dominate the landscape of proposed solutions. The diagram encapsulates the critical role of hybrid architectures and multi-objective optimization frameworks in achieving trade-offs between robustness, latency reduction, and overall system scalability. Additionally, the intricate connections between performance metrics — such as accuracy, privacy protection, and cost efficiency — underscore the inherent complexity of balancing computational demands with the evolving requirements of IIoT systems.

In Fig. 5, the emphasis is placed on addressing challenges related to networking and computational efficiency. The connections between problem statements such as heterogeneous data management, network optimization, and confidentiality vulnerabilities with methodologies like Federated Learning (FL), Deep Reinforcement Learning (DRL), and graph-based relation mining reveal the dominant reliance on intelligent decision-making algorithms for improving privacy, accuracy, and learning efficiency. Notable observations include methodologies like FedACS and RaFed achieving substantial accuracy improvements while simultaneously optimizing resource allocation, thereby underscoring the role of innovative frameworks in enhancing system reliability.

Together, these visualizations provides the broad spectrum of existing solutions and their contributions to advancing IIoT infrastructure, particularly in the context of FL and EC. They underscore the necessity of multi-disciplinary approaches to achieve optimal KPIs across privacy, efficiency, and learning accuracy. These findings collectively validate the effectiveness of integrating edge computing with federated learning, providing actionable insights into future research directions for IIoT applications.

6.4. Risk assessment and mitigation in FL for IIoT

FL and EC systems in IIoT face several attack vectors that can compromise data integrity, model accuracy, and system stability. Common threats include:

- **Model Poisoning Attacks:** Adversarial devices may inject incorrect gradients, thereby corrupting the global model. Defense mechanisms such as robust aggregation techniques, including auto-weighted geometric median and Byzantine fault-tolerant algorithms, are essential to mitigate these risks [26].
- **Inference Attacks:** Attackers may exploit gradient information to infer sensitive device data. Differential privacy and secure aggregation techniques help reduce inference risks, though they may introduce additional computational costs [38].
- **Eavesdropping and Data Leakage:** Data transmitted between edge devices and the server is vulnerable to eavesdropping. Homomorphic encryption and secure multi-party computation provide cryptographic solutions but require substantial resources, impacting real-time performance [24].

These security challenges necessitate adaptive strategies that balance privacy, efficiency, and resilience in IIoT environments.

6.5. Communication efficiency and model compression techniques

One of the primary challenges in implementing FL in IIoT environments is managing communication efficiency to minimize bandwidth use and latency. Techniques such as model compression, quantization, and sparsification have been explored to reduce the size of model updates without significantly impacting accuracy. For instance, gradient sparsification selectively transmits only the most informative gradients, thereby reducing data transfer needs. Additionally, structured pruning techniques remove redundant parameters, compressing the model for faster communication while maintaining performance. These methods are critical in IIoT, where network bandwidth is often limited, and transmission costs must be minimized to ensure system scalability.

FL involves frequent exchange of model parameters or gradients between edge devices and the central server. Reducing communication overhead is essential in constrained IIoT environments but can lead to challenges such as:

- **Delayed Model Updates:** Less frequent communication minimizes bandwidth usage but increases the risk of stale gradients, which can slow convergence.
- **Lossy Compression:** Techniques such as quantization and sparsification reduce communication costs but may discard critical information, resulting in suboptimal model updates.
- **Heterogeneous Resources:** Variations in computational and network capabilities among devices exacerbate the trade-off, particularly in non-IID data scenarios.

6.6. Optimization strategies

Several strategies have been proposed to address these trade-offs. For example:

- **Adaptive Communication:** Adjusting the frequency of updates dynamically based on model convergence rates to balance efficiency and accuracy.
- **Error-Compensated Compression:** Employing gradient sparsification or quantization with mechanisms to recover lost information, thereby maintaining accuracy while reducing communication costs.
- **Hierarchical Aggregation:** Introducing intermediate aggregation layers to reduce overall communication between edge devices and the central server, which is particularly useful in multi-layer IIoT architectures.

In a predictive maintenance [34,42,89] scenario for IIoT, real-time anomaly detection is critical. Frequent model updates ensure higher accuracy but may overwhelm bandwidth-constrained networks. In contrast, infrequent updates or lossy compression reduces communication cost but risks delayed or inaccurate anomaly detection. A hybrid approach that employs periodic full updates interspersed with compressed updates has been shown to strike a practical balance, maintaining both accuracy and efficiency.

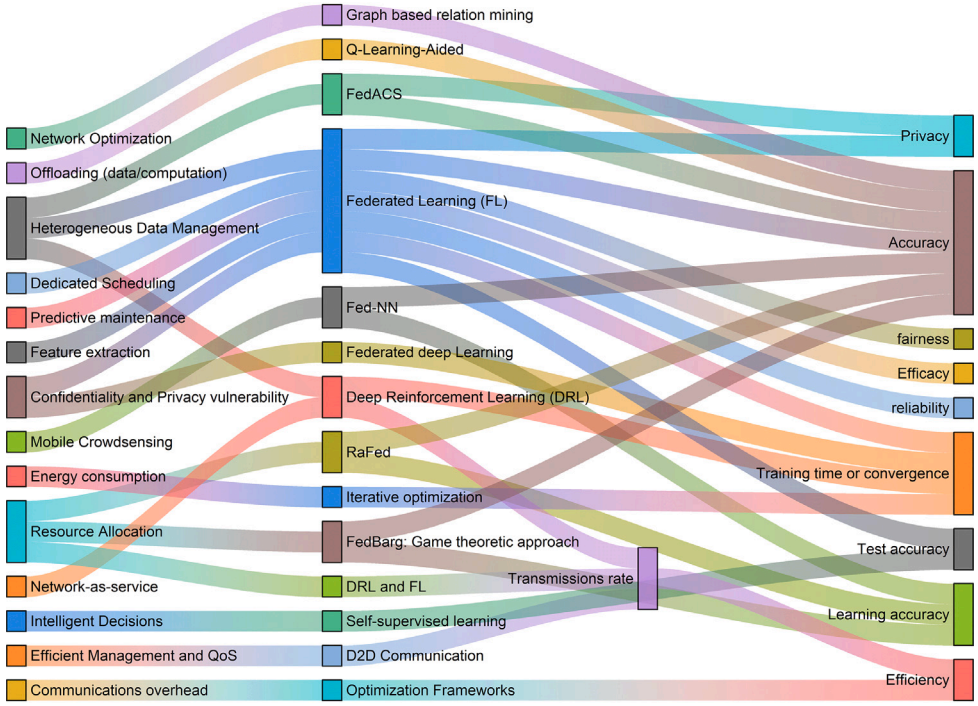


Fig. 5. KPI analysis based on the observation finding in of Table 6.

6.7. Limitations of edge devices in FL for IIoT

Edge devices in IIoT, such as microcontrollers and industrial sensors, often lack the processing power required for large-scale model training due to limited CPU, memory, and energy resources. These constraints make it challenging to perform complex computations, especially when training deep learning models [24]. Techniques such as model pruning, quantization, and federated distillation are increasingly employed to compress models and reduce computational demands, allowing for FL implementation even on resource-constrained edge devices. However, balancing model accuracy with reduced computational load remains an ongoing challenge in the field. The limited hardware capabilities of IIoT nodes, including constrained memory and processing power, restrict the use of deep learning models that require significant computational resources. As a result, lightweight neural networks and model optimization techniques, such as MobileNet and federated pruning, are often used to make FL feasible in these environments [47]. Additionally, distributed training approaches that leverage edge-cloud collaboration can offload intensive computations to cloud servers, allowing resource-limited devices to participate in FL without fully bearing the computational load.

6.8. Personalized federated learning for heterogeneous IIoT environments

Personalized federated learning (PFL) approaches are particularly suited to heterogeneous IIoT settings, where devices operate under varied conditions and data distributions. PFL enables each device to fine-tune the global model locally, optimizing performance for its specific context. Methods like meta-learning-based PFL and cluster-based aggregation allow each device to train a model that better fits its unique operating environment while still contributing to the global model [42]. This approach helps maintain high model accuracy across diverse devices, particularly in applications such as predictive maintenance where device-specific data is essential.

6.9. Synchronous vs. Asynchronous FL in IIoT systems

In IIoT, the choice between synchronous and asynchronous FL depends on latency requirements and device availability. Synchronous FL requires all devices to complete updates before aggregation, ensuring consistency but introducing latency, especially in resource-constrained devices. Asynchronous FL allows devices to update the global model as soon as they complete their local training, which reduces waiting times and improves efficiency in heterogeneous environments [47]. However, asynchronous methods can lead to model inconsistency, particularly if devices with outdated models contribute updates. Asynchronous FL is often preferable for IIoT applications that prioritize real-time responsiveness, while synchronous methods suit environments where model consistency is critical.

6.10. Applications of FL in IIoT: Beyond security

Beyond security, FL has significant applications in IIoT, particularly in predictive maintenance, quality control, and supply chain optimization:

- **Predictive Maintenance:** FL allows devices to collaboratively train on real-time sensor data from distributed equipment, enabling early fault detection and predictive analytics. This reduces downtime and maintenance costs by allowing for proactive repairs.
- **Quality Control:** FL facilitates real-time quality monitoring across multiple manufacturing units by aggregating defect detection models. This ensures consistent quality without centralizing sensitive production data.
- **Supply Chain Optimization:** FL enables secure and collaborative forecasting and inventory management across the supply chain. By aggregating data from multiple sources, FL can improve demand forecasting accuracy and reduce stockouts.

These applications demonstrate the versatility of FL in enhancing operational efficiency, safety, and resilience in IIoT systems.

6.11. Application contexts and limitations of FL frameworks in IIoT

Various FL frameworks are suitable for specific IIoT use cases, depending on the operational context, device heterogeneity, and data privacy requirements. For instance:

- **Hierarchical Federated Learning (HFL):** HFL is effective in large-scale IIoT deployments where devices are grouped by proximity to reduce communication overhead. However, HFL can suffer from latency due to multi-level aggregation, making it less suitable for real-time applications [21].
- **Asynchronous Federated Learning (AFL):** AFL is ideal for environments where devices have varied connectivity and computational resources. By allowing updates asynchronously, AFL minimizes delays from stragglers but may experience model drift in highly dynamic IIoT systems [47].
- **Personalized Federated Learning (PFL):** PFL supports heterogeneous device requirements by allowing customized model updates. It is beneficial in IIoT settings with diverse data distributions; however, personalization can compromise global model performance and increase computational overhead on constrained devices [61].

These frameworks highlight the need for IIoT-specific adaptations of FL to address limitations based on application context.

7. Conclusion

In conclusion, this work has conducted an in-depth exploration of the integration of FL and EC within IIoT systems, examining their combined potential and identifying the unique challenges that arise in this context. Key areas, such as privacy preservation, communication efficiency, resource constraints, and the complexities of distributed model training, were analyzed in detail, shows that it requires adaptive, scalable solutions. While these challenges highlight the complexity of deploying FL and EC in IIoT, they also underscore the opportunities for innovation, particularly in areas like adaptive model tuning, lightweight encryption techniques, and optimized aggregation protocols. The comprehensive review of existing frameworks and methodologies has illuminated the strengths and limitations of current approaches, with an evaluation based on critical performance metrics, including accuracy, latency, and energy efficiency. This assessment provides a nuanced understanding of how FL and EC can be practically and efficiently implemented in IIoT environments. The identified challenges necessitate further research into adaptive, context-specific frameworks that can handle the inherent heterogeneity of IIoT devices and applications. The work demonstrates that FL and EC are promising technology for the future of IIoT, enabling scalable, privacy-preserving, and efficient solutions. Addressing these challenges through innovative research will be essential in unlocking the full potential of IIoT, thereby paving the way for advanced applications in predictive maintenance, real-time monitoring, and autonomous industrial operations. The ongoing development in this field represents a critical pathway towards smarter, more secure, and highly adaptive IIoT systems.

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CRediT authorship contribution statement

Dinesh kumar sah: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Maryam Vahabi:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Hossein Fotouhi:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

There is no competing interests for this article.

Data availability

No data was used for the research described in the article.

References

[1] Wei Xian, Kan Yu, Fengling Han, Le Fang, Dehua He, Qing-Long Han, Advanced manufacturing in industry 5.0: A survey of key enabling technologies and future trends, IEEE Trans. Ind. Informat. (2023).
[2] Hao Ran Chi, Chung Kit Wu, Nen-Fu Huang, Kim Fung Tsang, Ayman Radwan, A survey of network automation for industrial internet-of-things towards industry 5.0, IEEE Trans. Ind. Informat. (2022).
[3] Tie Qiu, Jiancheng Chi, Xiaobo Zhou, Zhaolong Ning, Mohammed Atiquzzaman, Dapeng Oliver Wu, Edge computing in industrial internet of things: Architecture, advances and challenges, IEEE Commun. Surv. Tutor. 22 (4) (2020) 2462–2488.
[4] Qinbin Li, Zeyi Wen, Zhaomin Wu, Sixu Hu, Naibo Wang, Yuan Li, Xu Liu, Bingsheng He, A survey on federated learning systems: vision, hype and reality for data privacy and protection, IEEE Trans. Knowl. Data Eng. (2021).
[5] Viraaji Mothukuri, Reza M. Parizi, Seyedamin Pouriyeh, Yan Huang, Ali Dehghantanha, Gautam Srivastava, A survey on security and privacy of federated learning, Future Gener. Comput. Syst. 115 (2021) 619–640.
[6] Wei Yu, Fan Liang, Xiaofei He, William Grant Hatcher, Chao Lu, Jie Lin, Xinyu Yang, A survey on the edge computing for the internet of things, IEEE Access 6 (2017) 6900–6919.
[7] Tian Li, Anit Kumar Sahu, Ameet Talwalkar, Virginia Smith, Federated learning: Challenges, methods, and future directions, IEEE Signal Process. Mag. 37 (3) (2020) 50–60.
[8] Qiang Yang, Yang Liu, Tianjian Chen, Yongxin Tong, Federated machine learning: Concept and applications, ACM Trans. Intell. Syst. Technol. (TIST) 10 (2) (2019) 1–19.
[9] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, Yuan Gao, A survey on federated learning, Knowl.-Based Syst. 216 (2021) 106775.
[10] Mahadev Satyanarayanan, The emergence of edge computing, Computer 50 (1) (2017) 30–39.
[11] Wazir Zada Khan, Ejaz Ahmed, Saqib Hakak, Ibrar Yaqoob, Arif Ahmed, Edge computing: A survey, Future Gener. Comput. Syst. 97 (2019) 219–235.
[12] Qi Xia, Winson Ye, Zeyi Tao, Jindi Wu, Qun Li, A survey of federated learning for edge computing: Research problems and solutions, High- Confid. Comput. 1 (1) (2021) 100008.
[13] Fang Liu, Guoming Tang, Youhuizi Li, Zhiping Cai, Xingzhou Zhang, Tongqing Zhou, A survey on edge computing systems and tools, Proc. IEEE 107 (8) (2019) 1537–1562.
[14] Qi Guo, Fengxiao Tang, Nei Kato, Federated reinforcement learning-based resource allocation for D2D-aided digital twin edge networks in 6G industrial IoT, IEEE Trans. Ind. Informat. 19 (5) (2023) 7228–7236.
[15] Chenhan Zhang, Lei Cui, Shui Yu, James J.Q. Yu, A communication-efficient federated learning scheme for IoT-based traffic forecasting, IEEE Internet Things J. 9 (14) (2022) 11918–11931.
[16] Arash Heidari, Nima Jafari Navimipour, Mohammad Ali Jabraeil Jamali, Shahin Akbarpour, A green, secure, and deep intelligent method for dynamic IoT-edge-cloud offloading scenarios, Sustain. Computing: Informatics Syst. 38 (2023) 100859.
[17] Komeil Moghaddasi, Shakiba Rajabi, Farhad Soleimanian Gharehchopogh, Ali Ghaffari, An advanced deep reinforcement learning algorithm for three-layer D2D-edge-cloud computing architecture for efficient task offloading in the internet of things, Sustain. Comput. Informat. Syst. 43 (2024) 100992.
[18] Amir Pakmehr, Majid Gholipour, Esmaeil Zeinali, ETFC: Energy-efficient and deadline-aware task scheduling in fog computing, Sustain. Comput. Informat. Syst. 43 (2024) 100988.
[19] Mario Bochicchio, Sileshi Nibret Zeleke, Personalized federated learning in edge-cloud continuum for privacy-preserving health informatics: Opportunities and challenges, in: International Conference on Advanced Information Networking and Applications, Springer, 2024, pp. 368–378.

[20] Zhaoyang Wang, Song Wang, Zhiyao Zhao, Muyi Sun, Trustworthy localization with EM-based federated control scheme for iIoTs, IEEE Trans. Ind. Informat. 19 (1) (2023) 1069–1079.

[21] Xingjie Zeng, Zepei Yu, Weishan Zhang, Xiao Wang, Qinghua Lu, Tao Wang, Mu Gu, Yonglin Tian, Fei-Yue Wang, Homophily learning-based federated intelligence: A case study on industrial IoT equipment failure prediction, IEEE Internet Things J. 10 (8) (2023) 7356–7365.

[22] Chao Zhao, Weiming Shen, Federated domain generalization: A secure and robust framework for intelligent fault diagnosis, IEEE Trans. Ind. Informat. (2023) 1–9.

[23] Partha Pratim Ray, Dinesh Dash, Debashis De, Edge computing for internet of things: A survey, e-healthcare case study and future direction, J. Netw. Comput. Appl. 140 (2019) 1–22.

[24] Shunpu Tang, Lunyuan Chen, Ke He, Junjuan Xia, Lisheng Fan, Arumugam Nallanathan, Computational intelligence and deep learning for next-generation edge-enabled industrial IoT, IEEE Trans. Netw. Sci. Eng. 10 (5) (2023) 2881–2893.

[25] Tantan Zhao, Fan Li, Lijun He, DRL-based joint resource allocation and device orchestration for hierarchical federated learning in NOMA-enabled industrial IoT, IEEE Trans. Ind. Informat. 19 (6) (2023) 7468–7479.

[26] Shenghui Li, Edith Ngai, Thiemo Voigt, Byzantine-robust aggregation in federated learning empowered industrial IoT, IEEE Trans. Ind. Informat. 19 (2) (2023) 1165–1175.

[27] Teng Ma, Haibo Wang, Chong Li, Quantized distributed federated learning for industrial internet of things, IEEE Internet Things J. 10 (4) (2023) 3027–3036.

[28] Made Adi Paramartha Putra, Adinda Riztia Putri, Ahmad Zainudin, Dong-Seong Kim, Jae-Min Lee, ACS: Accuracy-based client selection mechanism for federated industrial IoT, Internet Things (Netherlands) 21 (2023) All Open Access, Bronze Open Access.

[29] Xiaolong Xu, Haoyuan Li, Zheng Li, Xiaokang Zhou, Safe: Synergic data filtering for federated learning in cloud-edge computing, IEEE Trans. Ind. Informat. 19 (2) (2023) 1655–1665.

[30] Bahar Farahani, Amin Karimi Monsefi, Smart and collaborative industrial IoT: A federated learning and data space approach, Digit. Commun. Netw. (2023).

[31] Wei Yang Bryan Lim, Zehui Xiong, Jiawen Kang, Dusit Niyato, Cyril Leung, Chunyan Miao, Xuemin Shen, When information freshness meets service latency in federated learning: A task-aware incentive scheme for smart industries, IEEE Trans. Ind. Informat. 18 (1) (2022) 457–466.

[32] Qi Feng, Debiao He, Jian Shen, Min Luo, Kim-Kwang Raymond Choo, PpNNT: Multi-party privacy-preserving neural network training system, IEEE Trans. Artif. Intell. (2023) 1–14.

[33] Bo Xu, Haitao Zhao, Haotong Cao, Sahil Garg, Georges Kaddoum, Mohammad Mehedi Hassan, Edge aggregation placement for semi-decentralized federated learning in industrial internet of things, Future Gener. Comput. Syst. 150 (2024) 160–170.

[34] Y.P. Tsang, C.H. Wu, W.H. Ip, C.K.M. Lee, Federated-learning-based decision support for industrial internet of things (iIoT)-based printed circuit board assembly process, J. Grid Comput. 20 (4) (2022).

[35] Jiao Chen, Jianhua Tang, Weihua Li, Industrial edge intelligence: Federated-meta learning framework for few-shot fault diagnosis, IEEE Trans. Netw. Sci. Eng. 10 (6) (2023) 3561–3573.

[36] Shijing Yuan, Beiyu Dong, Hongtao Lvy, Hongze Liu, Hongyang Chen, Chentao Wu, Song Guo, Yue Ding, Jie Li, Adaptive incentivize for cross-silo federated learning in iIoT: A multi-agent reinforcement learning approach, IEEE Internet Things J. (2023) 1.

[37] Xiuzhao Ji, Jie Tian, Haixia Zhang, Dalei Wu, Tiantian Li, Joint device selection and bandwidth allocation for cost-efficient federated learning in industrial internet of things, IEEE Internet Things J. 10 (10) (2023) 9148–9160.

[38] Laizhong Cui, Jiating Ma, Yipeng Zhou, Shui Yu, Boosting accuracy of differentially private federated learning in industrial IoT with sparse responses, IEEE Trans. Ind. Informat. 19 (1) (2023) 910–920.

[39] Chao Ma, Haiyang Yu, Zheng Li, Zhen Yang, Federated learning framework based on data value evaluation in industrial IoT, Secur. Commun. Netw. 2022 (2022).

[40] Truong Thu Huong, Ta Phuong Bac, Dao Minh Long, Tran Duc Luong, Nguyen Minh Dan, Le Anh Quang, Le Thanh Cong, Bui Doan Thang, Kim Phuc Tran, Detecting cyberattacks using anomaly detection in industrial control systems: A federated learning approach, Comput. Ind. 132 (2021).

[41] Huong Thu Truong, Bac Phuong Ta, Quang Anh Le, Dan Minh Nguyen, Cong Thanh Le, Hoang Xuan Nguyen, Ha Thu Do, Hung Tai Nguyen, Kim Phuc Tran, Light-weight federated learning-based anomaly detection for time-series data in industrial control systems, Comput. Ind. 140 (2022) 103692.

[42] Yuange Liu, Zhicheng Bao, Yuqian Wang, Xingjie Zeng, Liang Xu, Weishan Zhang, Hongwei Zhao, Zepei Yu, The adaptive personalized federated meta-learning for anomaly detection of industrial equipment, IEEE J. Radio Freq. Identif. 6 (2022) 832–836.

[43] Mohamed Abdel-Basset, Nour Moustafa, Hossam Hawash, Privacy-preserved generative network for trustworthy anomaly detection in smart grids: A federated semisupervised approach, IEEE Trans. Ind. Informat. 19 (1) (2023) 995–1005.

[44] Subir Halder, Thomas Newe, Radio fingerprinting for anomaly detection using federated learning in lora-enabled industrial internet of things, Future Gener. Comput. Syst. 143 (2023) 322–336, Hybrid Gold Open Access.

[45] Jianhua Li, Lingjuan Lyu, Ximeng Liu, Xuyun Zhang, Xixiang Lyu, FLEAM: A federated learning empowered architecture to mitigate DDoS in industrial IoT, IEEE Trans. Ind. Informat. 18 (6) (2022) 4059–4068.

[46] Xiaoding Wang, Sahil Garg, Hui Lin, Jia Hu, Georges Kaddoum, Md Jalil Piran, M. Shamim Hossain, Toward accurate anomaly detection in industrial internet of things using hierarchical federated learning, IEEE Internet Things J. 9 (10) (2022) 7110–7119.

[47] Chen Zhao, Zhipeng Gao, Qian Wang, Kaile Xiao, Zijia Mo, AFL: An adaptively federated multitask learning for model sharing in industrial IoT, IEEE Internet Things J. 9 (18) (2022) 17080–17088.

[48] Ons Aouedi, Kandaraj Piamrat, Guillaume Muller, Kamal Singh, Federated semisupervised learning for attack detection in industrial internet of things, IEEE Trans. Ind. Informat. 19 (1) (2023) 286–295.

[49] Mikail Mohammed Salim, Abir El Azzaoui, Xianjun Deng, Jong Hyuk Park, FL-CTIF: A federated learning based CTI framework based on information fusion for secure iIoT, Inf. Fusion 102 (2024).

[50] Priyanka Verma, John G. Breslin, Donna O'Shea, FLDID: Federated learning enabled deep intrusion detection in smart manufacturing industries, Sensors 22 (22) (2022) Green Open Access.

[51] Shafkat Islam, Shahriar Badsha, Shamik Sengupta, Ibrahim Khalil, Mohammed Atiquzzaman, An intelligent privacy preservation scheme for EV charging infrastructure, IEEE Trans. Ind. Informat. 19 (2) (2023) 1238–1247.

[52] Yunfei Song, Tian Liu, Tongquan Wei, Xiangfeng Wang, Zhe Tao, Mingsong Chen, FDA3: Federated defense against adversarial attacks for cloud-based iIoT applications, IEEE Trans. Ind. Informat. 17 (11) (2021) 7830–7838.

[53] Rahim Taheri, Mohammad Shojafar, Mamoun Alazab, Rahim Tafazolli, FED-iIoT: A robust federated malware detection architecture in industrial IoT, IEEE Trans. Ind. Informat. (2020).

[54] Muna Al-Hawawreh, Elena Sitnikova, Neda Aboutorab, Asynchronous peer-to-peer federated capability-based targeted ransomware detection model for industrial IoT, IEEE Access 9 (2021) 148738–148755.

[55] Yudian Huang, Meng Li, F. Richard Yu, Pengbo Si, Yanhua Zhang, Performance optimization for energy-efficient industrial internet of things based on ambient backscatter communication: An A3C-FL approach, IEEE Trans. Green Commun. Netw. 7 (3) (2023) 1121–1134.

[56] Manan Mehta, Siyuan Chen, Haichuan Tang, Chenhui Shao, A federated learning approach to mixed fault diagnosis in rotating machinery, J. Manuf. Syst. 68 (2023) 687–694.

[57] Venkatraman Balasubramanian, Moayad Aloqaily, Martin Reisslein, Fed-TSN: Joint failure probability-based federated learning for fault-tolerant time-sensitive networks, IEEE Trans. Netw. Serv. Manag. 20 (2) (2023) 1470–1486.

[58] Yingying Liang, Peng Zhao, Yimeng Wang, Federated few-shot learning-based machinery fault diagnosis in the industrial internet of things, Appl. Sci. (Switzerland) 13 (18) (2023).

[59] Mingkai Chen, Lindong Zhao, Jianxin Chen, Xin Wei, Mohsen Guizani, Modal-aware resource allocation for cross-modal collaborative communication in iIoT, IEEE Internet Things J. 10 (17) (2023) 14952–14964.

[60] Mengxuan Du, Haifeng Zheng, Xinxin Feng, Youjia Chen, Tiesong Zhao, Decentralized federated learning with Markov chain based consensus for industrial IoT networks, IEEE Trans. Ind. Informat. 19 (4) (2023) 6006–6015.

[61] Jianfeng Lu, Haibo Liu, Riheng Jia, Jiangtao Wang, Lichao Sun, Shaohua Wan, Toward personalized federated learning via group collaboration in iIoT, IEEE Trans. Ind. Informat. 19 (8) (2023) 8923–8932.

[62] Xiumei Deng, Jun Li, Chuan Ma, Kang Wei, Long Shi, Ming Ding, Wen Chen, Low-latency federated learning with DNN partition in distributed industrial IoT networks, IEEE J. Sel. Areas Commun. 41 (3) (2023) 755–775.

[63] Peiying Zhang, Chao Wang, Chunxiao Jiang, Zhu Han, Deep reinforcement learning assisted federated learning algorithm for data management of IIoT, IEEE Trans. Ind. Informat. 17 (2) (2021) 8475–8484.

[64] Zonghang Li, Yihong He, Fedgs: A federated group synchronization framework for heterogeneous data[formula presented], Softw. Impacts 12 (2022).

[65] Zibo Wang, Yifei Zhu, Dan Wang, Zhu Han, Federated analytics informed distributed industrial IoT learning with non-IID data, IEEE Trans. Netw. Sci. Eng. 10 (5) (2023) 2924–2939.

[66] Suiyuan Wu, Hongmei Xue, Long Zhang, Q-learning-aided offloading strategy in edge-assisted federated learning over industrial IoT, Electron. (Switzerland) 12 (7) (2023).

[67] Aishwariya Chakraborty, Sudip Misra, QoS-aware resource bargaining for federated learning over edge networks in industrial IoT, IEEE Trans. Netw. Sci. Eng. 10 (5) (2023) 2769–2778.

[68] Beibei Li, Yaxin Shi, Qinglei Kong, Qingyun Du, Rongxing Lu, Incentive-based federated learning for digital-twin-driven industrial mobile crowdsensing, IEEE Internet Things J. 10 (20) (2023) 17851–17864.

[69] Veronika Stephanie, Ibrahim Khalil, Mohammed Atiquzzaman, Digital twin enabled asynchronous SplitFed learning in E-healthcare systems, IEEE J. Sel. Areas Commun. (2023) 1.

[70] Shiva Raj Pokhrel, Learning from data streams for automation and orchestration of 6G industrial IoT: toward a semantic communication framework, Neural Comput. Appl. 34 (18) (2022) 15197–15206, Hybrid Gold Open Access.

[71] Yung-Lin Hsu, Chen-Feng Liu, Hung-Yu Wei, Mehdi Bennis, Optimized data sampling and energy consumption in iIoT: A federated learning approach, IEEE Trans. Commun. 70 (12) (2022) 7915–7931.
[72] Zonghang Li, Yihong He, Hongfang Yu, Jiawen Kang, Xiaoping Li, Zenglin Xu, Dusit Niyato, Data heterogeneity-robust federated learning via group client selection in industrial IoT, IEEE Internet Things J. 9 (18) (2022) 17844–17857.
[73] Hui Yang, Jiaqi Yuan, Chao Li, Guanliang Zhao, Zhengjie Sun, Qiuyan Yao, Bowen Bao, Athanasios V. Vasilakos, Jie Zhang, BrainIoT: Brain-like productive services provisioning with federated learning in industrial IoT, IEEE Internet Things J. 9 (3) (2022) 2014–2024.
[74] Giampaolo Cuozzo, Jonas Pettersson, Massimo Condoluci, Analysis of a contention-based approach over 5G NR for federated learning in an industrial internet of things scenario, IEEE Access 11 (2023) 74473–74485, Green Open Access.
[75] Wei Xiang, Jie Li, Yuan Zhou, Peng Cheng, Jiong Jin, Kan Yu, Digital twin empowered industrial IoT based on credibility-weighted swarm learning, IEEE Trans. Ind. Informat. (2023) 1–10.
[76] Zheqi Zhu, Shuo Wan, Pingyi Fan, Khaled B. Letaief, Federated multiagent actor-critic learning for age sensitive mobile-edge computing, IEEE Internet Things J. 9 (2) (2022) 1053–1067.
[77] Maysaa Khalil, Moez Esseghir, Leila Merghem-Boulahia, A federated learning approach for thermal comfort management, Adv. Eng. Informat. 52 (2022).
[78] Hui Zhang, Revolutionizing digital marketing industrial internet of things-enabled supply chain management in smart manufacturing, Computer- Aided Des. Appl. 21 (S4) (2024) 211–228.
[79] Bouziane Brik, Mourad Messaadia, M'hammed Sahnoun, Belgacem Bettayeb, Mohamed Amin Benatia, Fog-supported low-latency monitoring of system disruptions in industry 4.0: A federated learning approach, ACM Trans. Cyber- Phys. Syst. 6 (2) (2022).
[80] Vandana Bharti, Abhinav Kumar, Vishal Purohit, Rishav Singh, Amit Kumar Singh, Sanjay Kumar Singh, A label efficient semi self-supervised learning framework for IoT devices in industrial process, IEEE Trans. Ind. Informat. (2023) 1–10.
[81] Dong Yang, Weiting Zhang, Qiang Ye, Chuan Zhang, Ning Zhang, Chuan Huang, Hongke Zhang, Xuemin Shen, DetFed: Dynamic resource scheduling for deterministic federated learning over time-sensitive networks, IEEE Trans. Mob. Comput. (2023) 1–17.
[82] Seifeddine Messaoud, Abbas Bradai, Olfa Ben Ahmed, Pham Tran Anh Quang, Mohamed Atri, M. Shamim Hossain, Deep federated Q-learning-based network slicing for industrial IoT, IEEE Trans. Ind. Informat. 17 (8) (2021) 5572–5582.
[83] Weifeng Gao, Zhiwei Zhao, Geyong Min, Qiang Ni, Yuhong Jiang, Resource allocation for latency-aware federated learning in industrial internet-of-things, IEEE Trans. Ind. Informat. (2021).
[84] Yuan Gao, Guangming Zhang, Chunchun Zhang, Jinke Wang, Laurence T. Yang, Yaliang Zhao, Federated tensor decomposition-based feature extraction approach for industrial IoT, IEEE Trans. Ind. Informat. 17 (12) (2021) 8541–8549.
[85] Yunlong Lu, Xiaohong Huang, Ke Zhang, Sabita Maharjan, Yan Zhang, Communication-efficient federated learning for digital twin edge networks in industrial IoT, IEEE Trans. Ind. Informat. 17 (8) (2021) 5709–5718.
[86] Yinghao Guo, Zichao Zhao, Ke He, Shiwei Lai, Junjuan Xia, Lisheng Fan, Efficient and flexible management for industrial internet of things: A federated learning approach, Comput. Netw. 192 (2021).
[87] Wen Sun, Shiyu Lei, Lu Wang, Zhiqiang Liu, Yan Zhang, Adaptive federated learning and digital twin for industrial internet of things, IEEE Trans. Ind. Informat. 17 (8) (2021) 5605–5614.
[88] Woonghee Lee, Federated reinforcement learning-based UAV swarm system for aerial remote sensing, Wirel. Commun. Mob. Comput. 2022 (2022).
[89] Taimur Hafeez, Lina Xu, Gavin McArdle, Edge intelligence for data handling and predictive maintenance in IIoT, IEEE Access 9 (2021) 49355–49371.
[90] Muhammad Ayat Hidayat, Yugo Nakamura, Yutaka Arakawa, Privacy-preserving federated learning with resource adaptive compression for edge devices, IEEE Internet Things J. (2023).
[91] Wending Liu, Rui Han, Xinyu Guo, Junyan Ouyang, Xiaojiang Zuo, Chi Harold Liu, Neurondp: Neuron-grained differential privacy of deep neural networks in edge-based federated learning, in: 2024 IEEE 4th International Conference on Electronic Technology, Communication and Information, ICETCI, IEEE, 2024, pp. 166–171.
[92] Jie Fu, Yuan Hong, Xinpeng Ling, Leixia Wang, Xun Ran, Zhiyu Sun, Wendy Hui Wang, Zhili Chen, Yang Cao, Differentially private federated learning: A systematic review, 2024, arXiv preprint arXiv:2405.08299.
[93] Ahmad Hammoud, Hadi Otrok, Azzam Mourad, Zbigniew Dziong, On demand fog federations for horizontal federated learning in IoV, IEEE Trans. Netw. Serv. Manag. 19 (3) (2022) 3062–3075.
[94] Lunyuan Chen, Lisheng Fan, Xianfu Lei, Trung Q. Duong, Arumugam Nallanathan, George K. Karagiannidis, Relay-assisted federated edge learning: Performance analysis and system optimization, IEEE Trans. Commun. (2023).
[95] Mohamad Arafeh, Hadi Otrok, Hakima Ould-Slimane, Azzam Mourad, Chamseddine Talhi, Ernesto Damiani, ModularFed: Leveraging modularity in FL frameworks, Internet Things 22 (2023) 100694.
[96] Aditya Pribadi Kalapaaking, Ibrahim Khalil, Mohammed Atiquzzaman, Smart policy control for securing federated learning management system, IEEE Trans. Netw. Serv. Manag. (2023).
[97] Zhenpeng Liu, Zilin Gao, Jingyi Wang, Qiannan Liu, Jianhang Wei, et al., Ppefl: An edge federated learning architecture with privacy-preserving mechanism, Wirel. Commun. Mob. Comput. 2022 (2022).
[98] Yunlong Lu, Xiaohong Huang, Yueyue Dai, Sabita Maharjan, Yan Zhang, Differentially private asynchronous federated learning for mobile edge computing in urban informatics, IEEE Trans. Ind. Informat. 16 (3) (2019) 2134–2143.